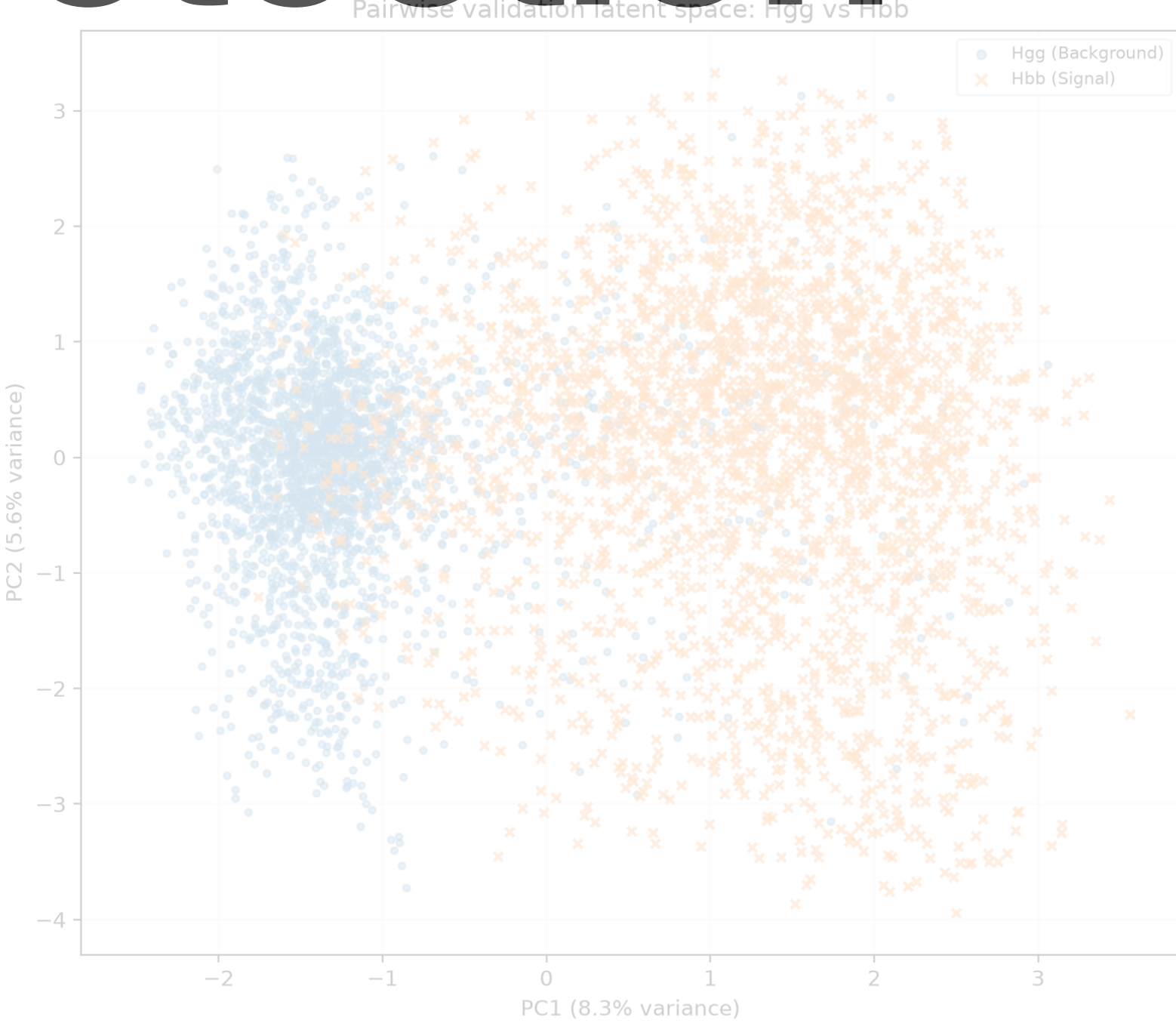
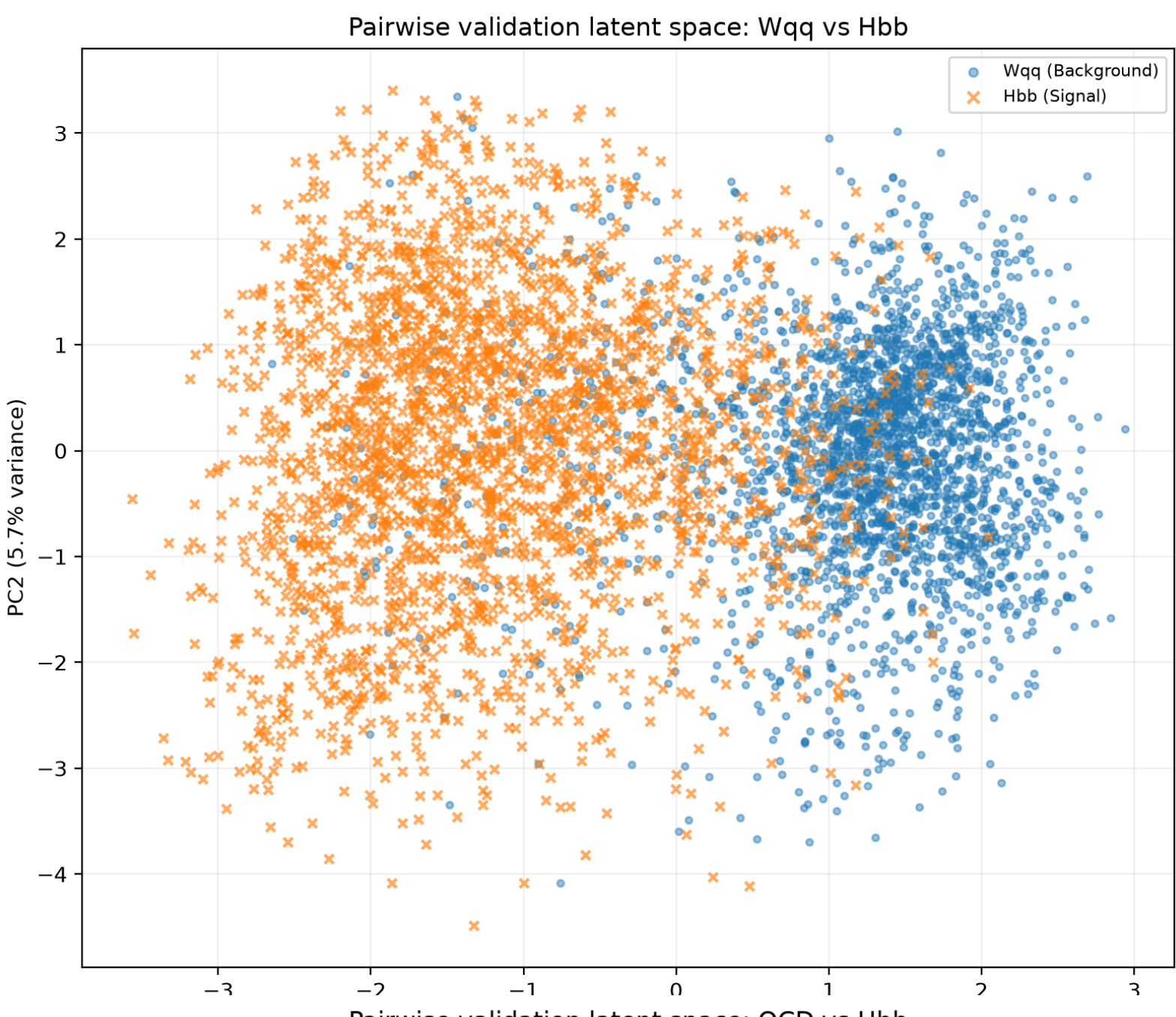


Multi-background Anomaly Detection

Lixing Wang, Hanlei Gao

Jul 14, 2026



Introduction

We continue to build on LeJEPAs representations for anomaly detection. We scale data size from a single QCD background to a multi-background JetClass setting, and find that the learned representation contains strong pairwise anomaly information.

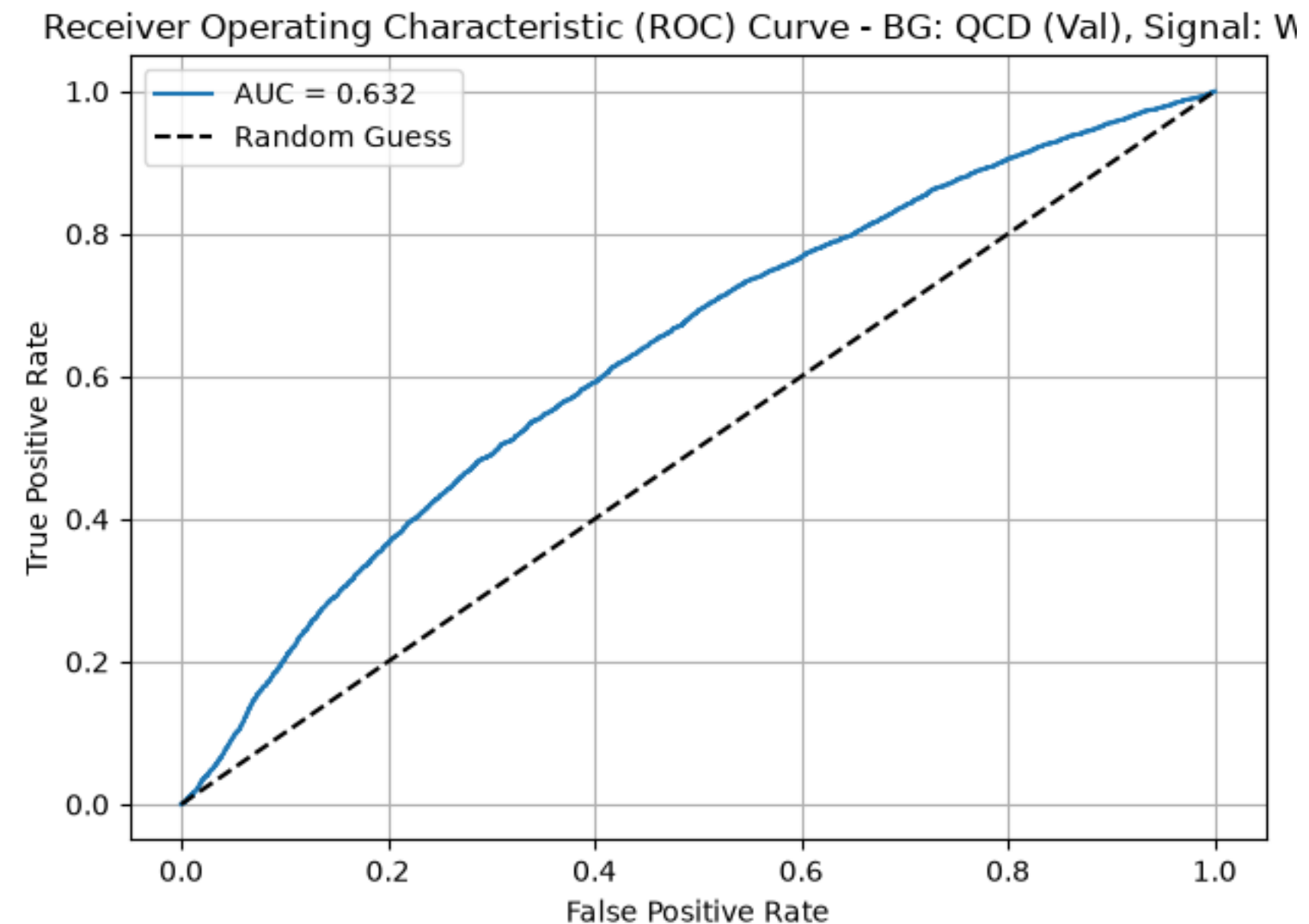
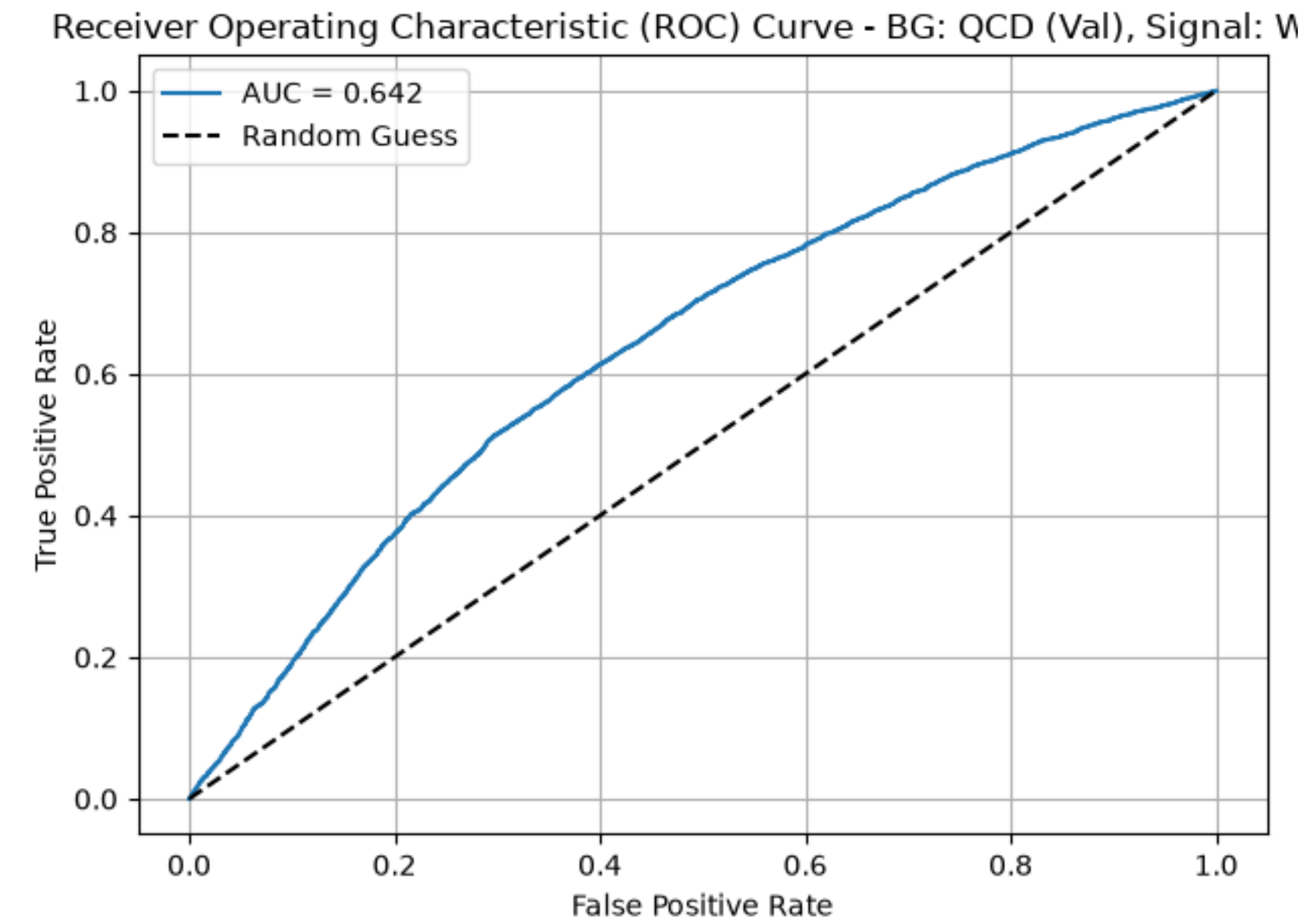
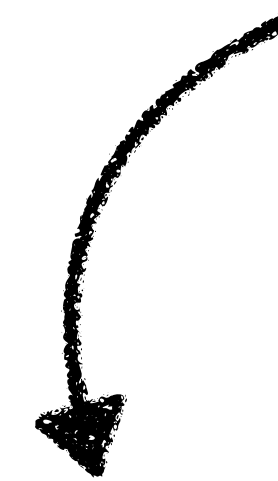
Motivation

- Single-background anomaly detection gives a clean evaluation protocol: train on one dominant background, then test whether a signal class lies outside the learned normal region.
- But in the multi-background case, the “normal” region is no longer a single compact distribution. It is a mixture of several physically different jet types.
- **Key observation**
- A global all-background Mahalanobis score can look almost random, while per-class Mahalanobis scores reveal strong separation.
- For example, in the Hbb-as-signal model, the merged-background Mahalanobis AUC is only **0.539**, but the per-class scores reach **0.928** for QCD vs Hbb, **0.846** for Wqq vs Hbb, and **0.818** for Hgg vs Hbb.

Single-background Baseline

What we fixed

- Fixed a Mahalanobis-score bug: the previous implementation only used the diagonal covariance terms.
- After using the full covariance matrix, QCD vs Wqq AUC changed from **0.642** -> **0.632**
- This gives a more conservative but more accurate single-background baseline



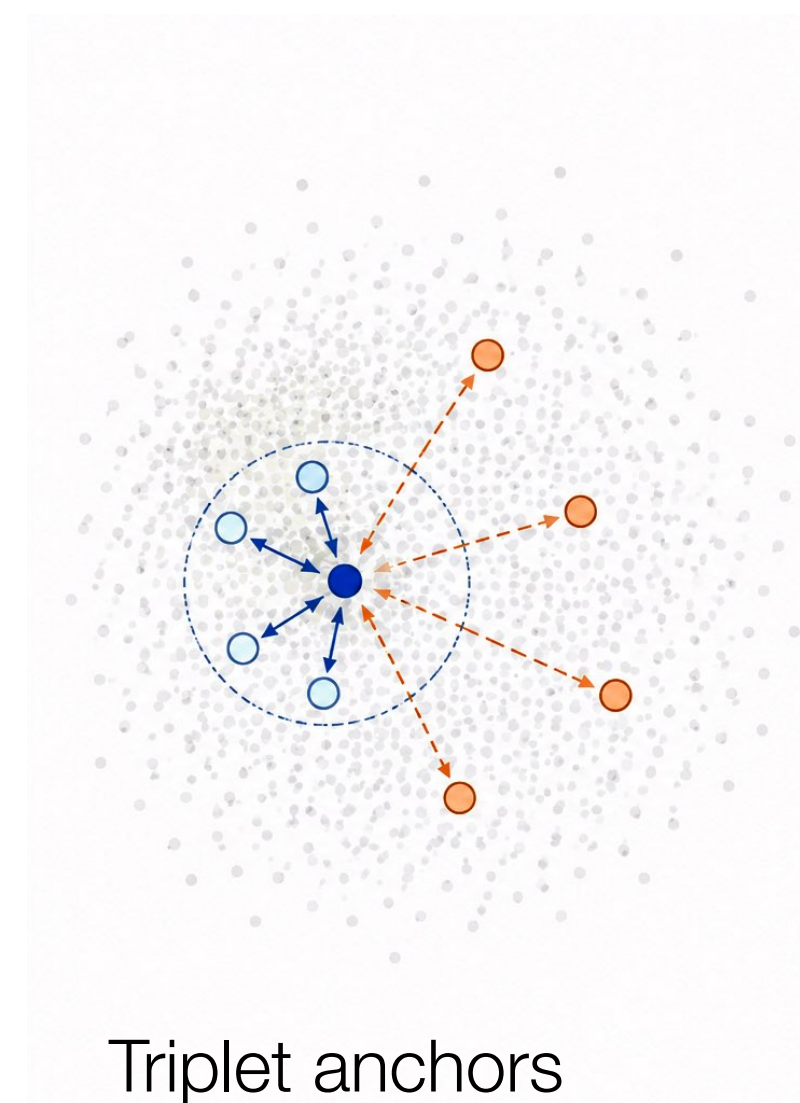
What we learned from single-background QCD vs Wqq

- Batch mixing must be conservative.
- Randomizing anchor fraction too broadly, e.g. **0.1-0.9**, collapsed AUC to about **0.53**.
- pt-drop views should not be too aggressive.
- Global **0.0-0.3** and local **0.3-0.75** outperformed the more aggressive **0.0-0.5 \ 0.5-0.95** split by around **0.05 AUC**.
- Triplet margin should not be too large.
- Margin **0.5** was better than **1.0** or **2.0**, likely because large margins let the triplet objective dominate for too long and encourage overfitting to artificial augmentation patterns.

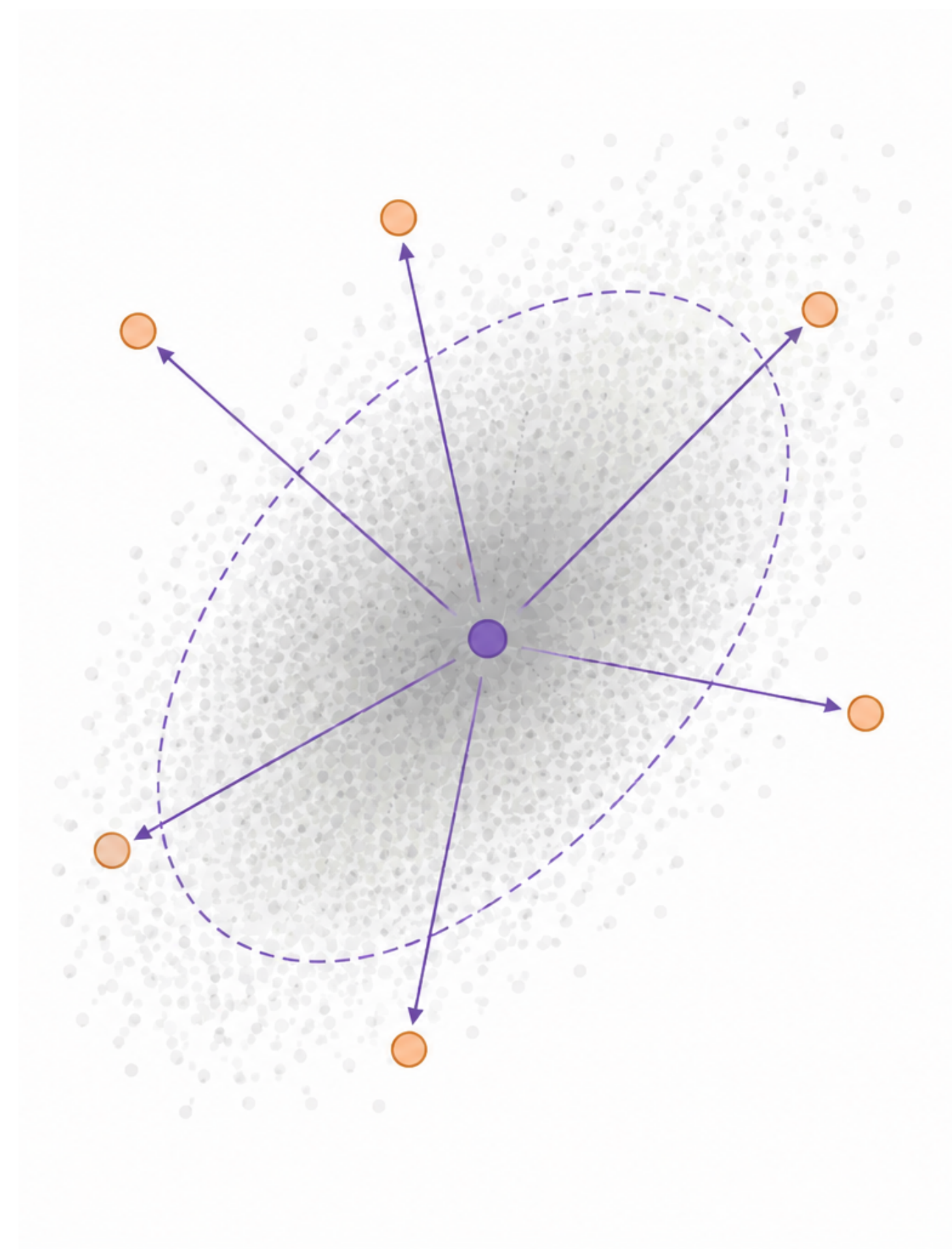
Failed but useful experiment: Mahalanobis loss

A negative result that clarified the role of triplet loss

- Triplet loss anchors each event's negative views against its own positive/global view.
- We tested a different idea: push all negative views away from the EMA mean distribution of **all** training positive views using a Mahalanobis loss.



Triplet anchors to an event.

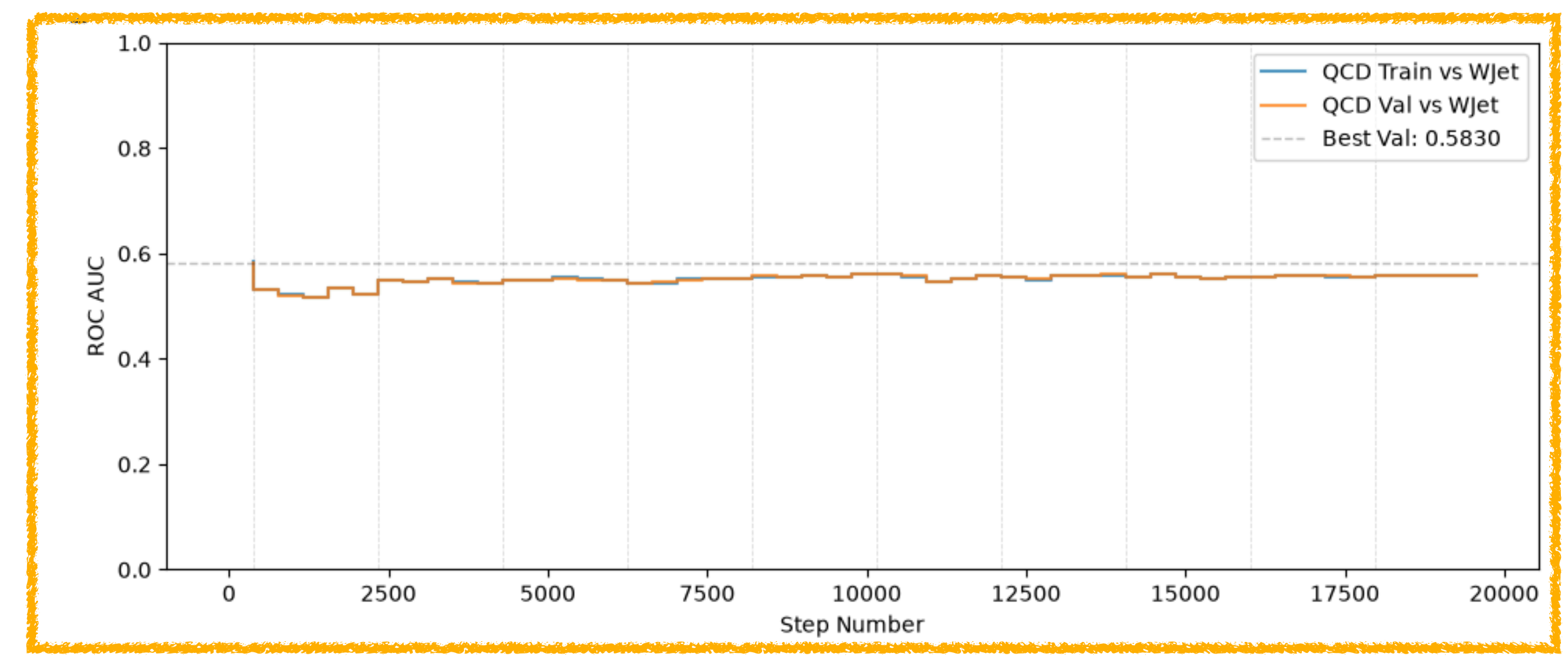
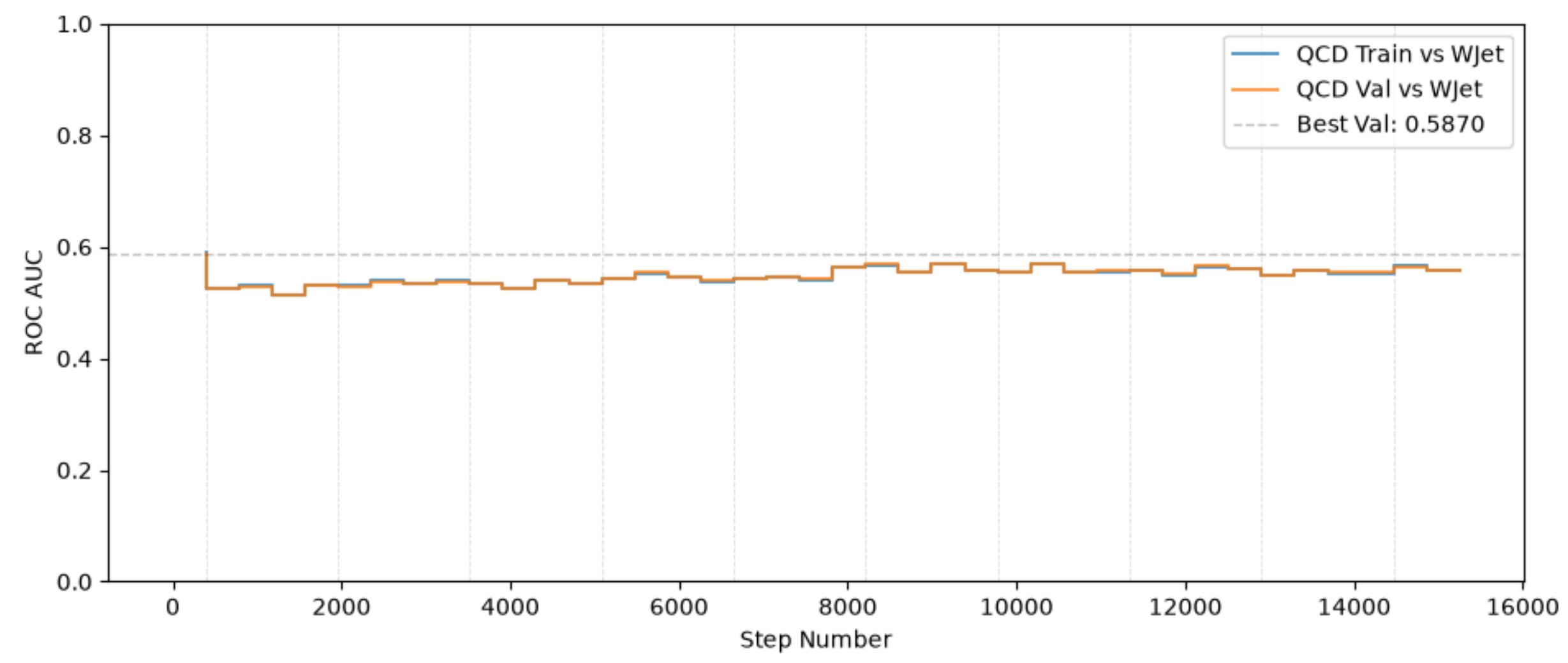
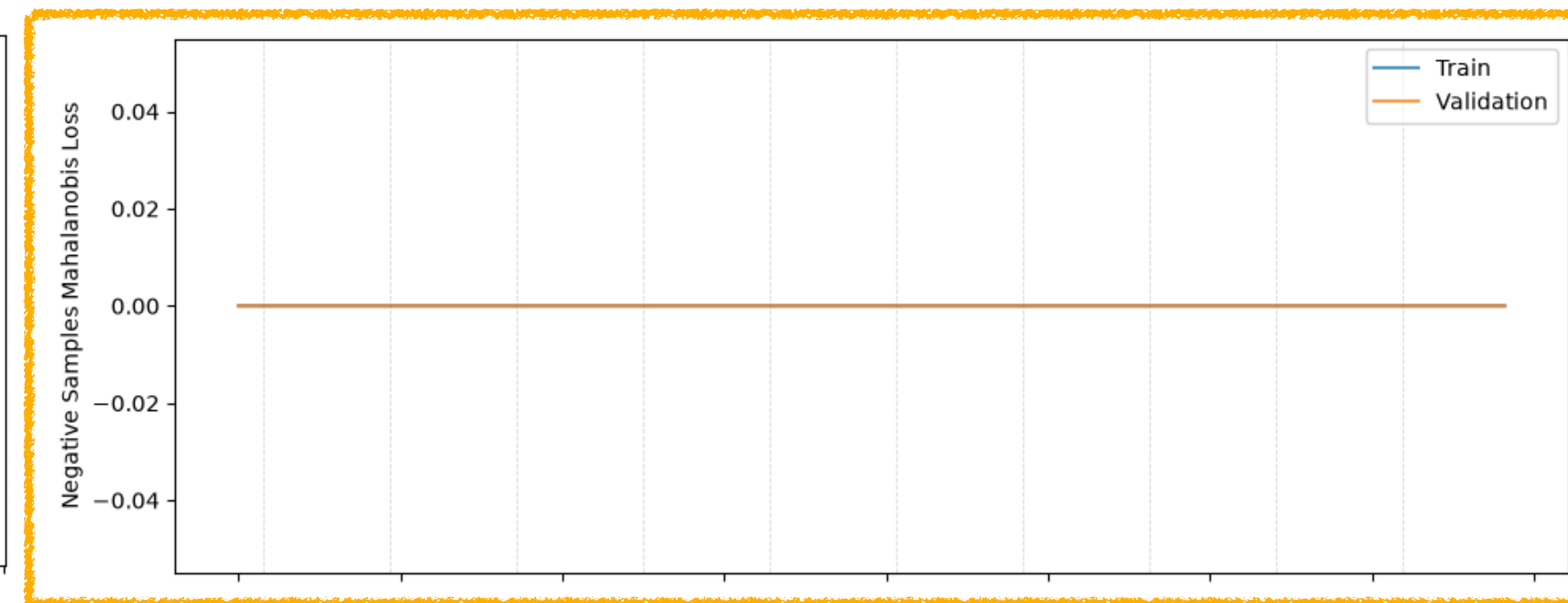
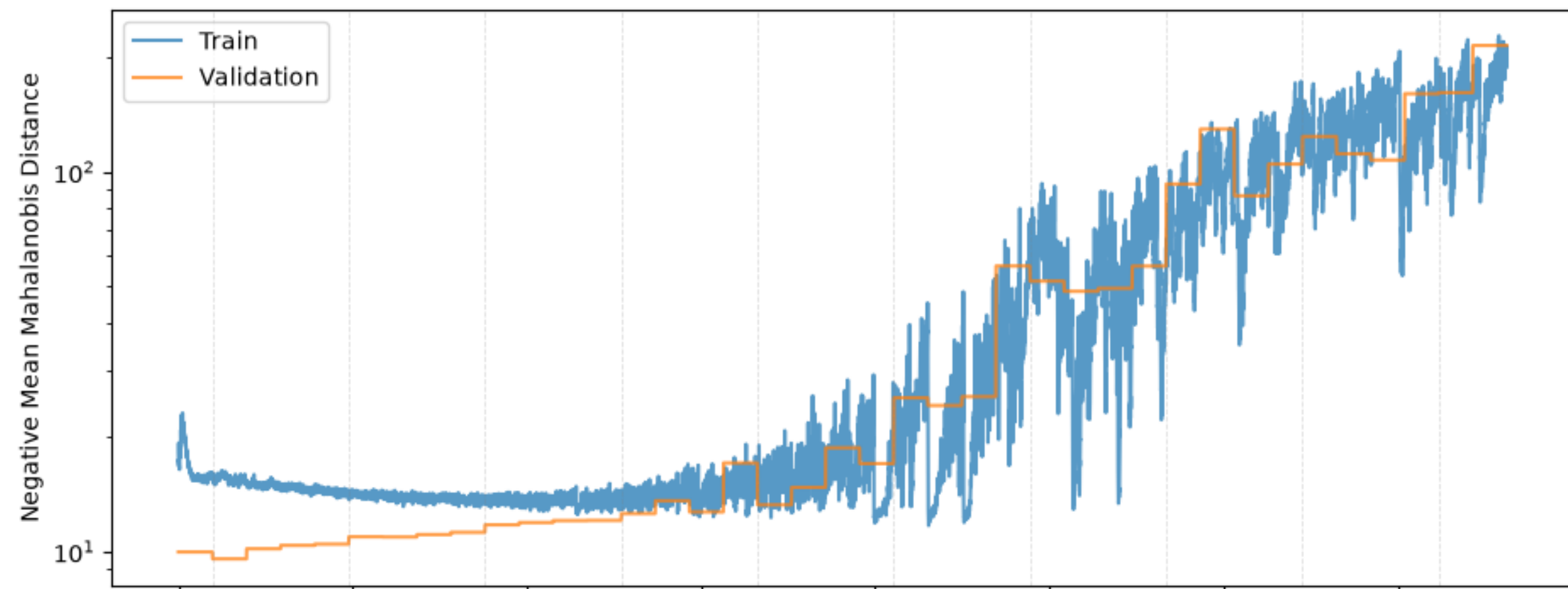


Mahalanobis loss anchors to EMA of all train positives..

Failed but useful experiment: Mahalanobis loss

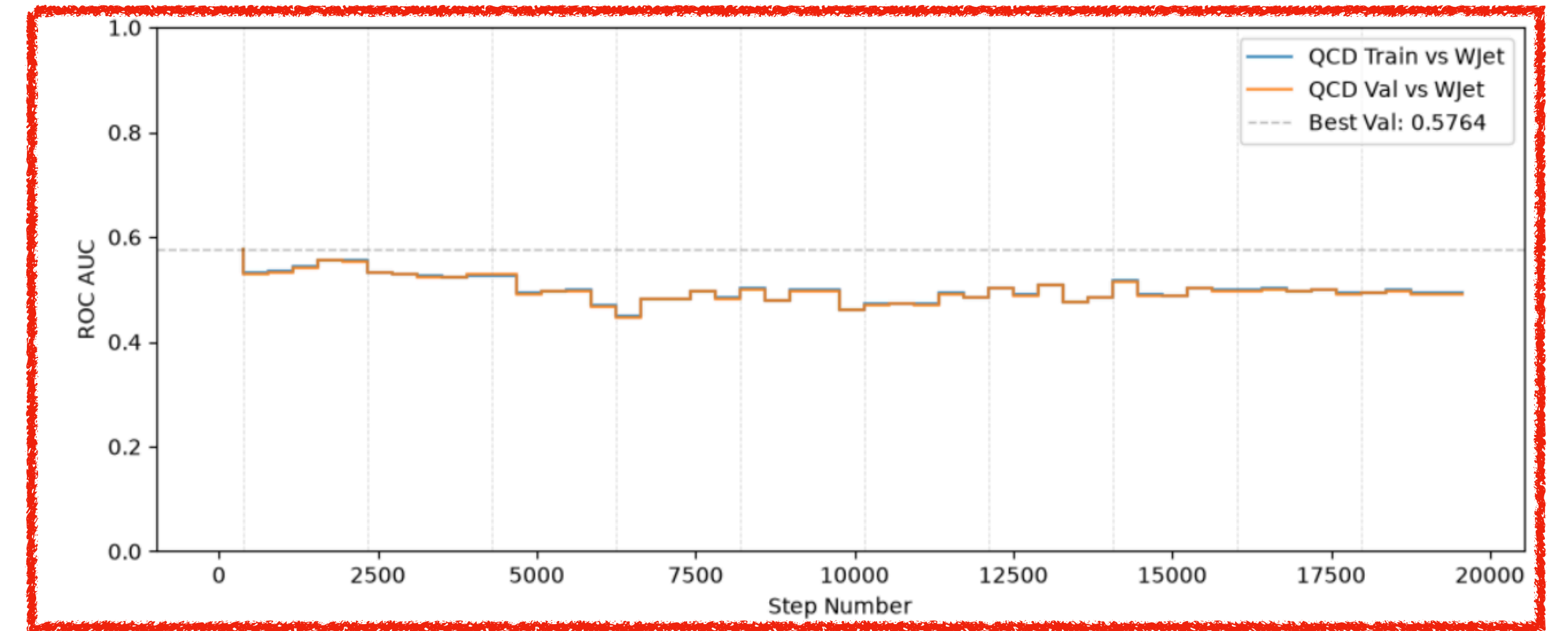
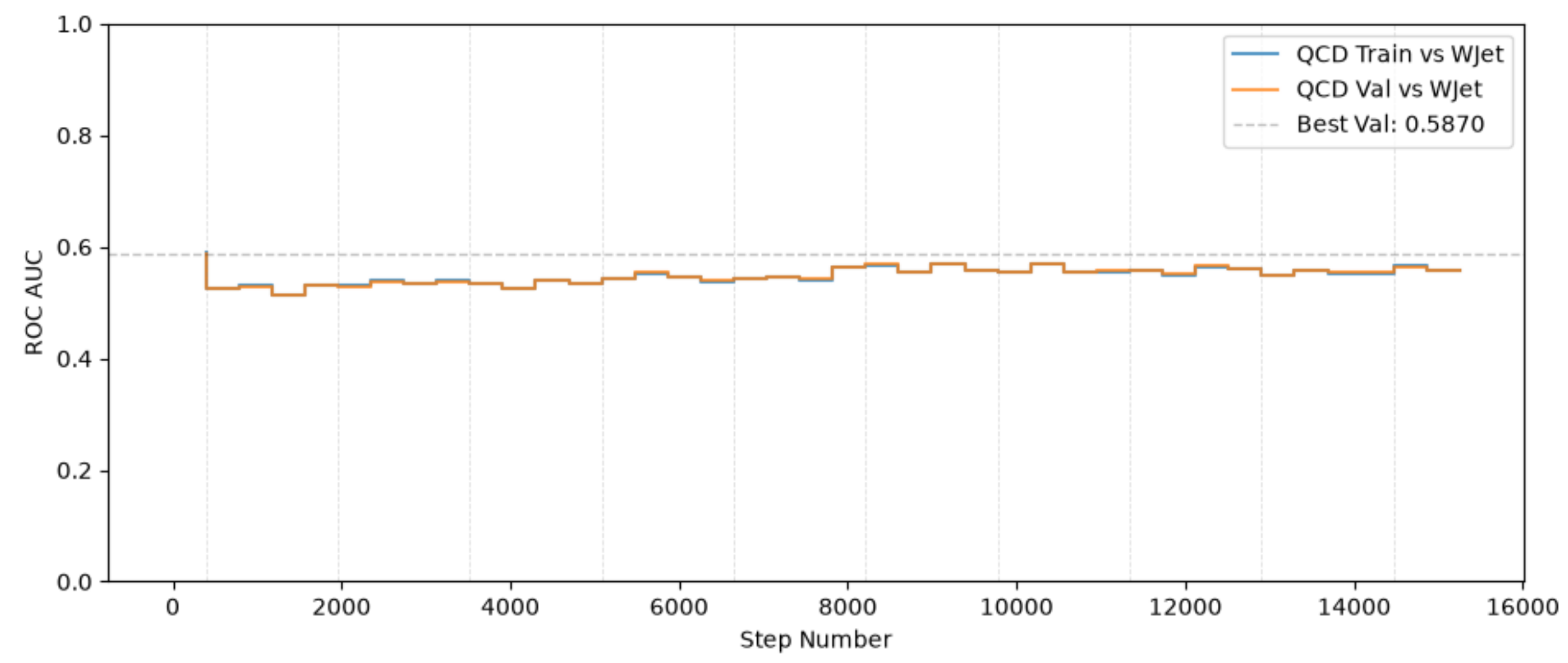
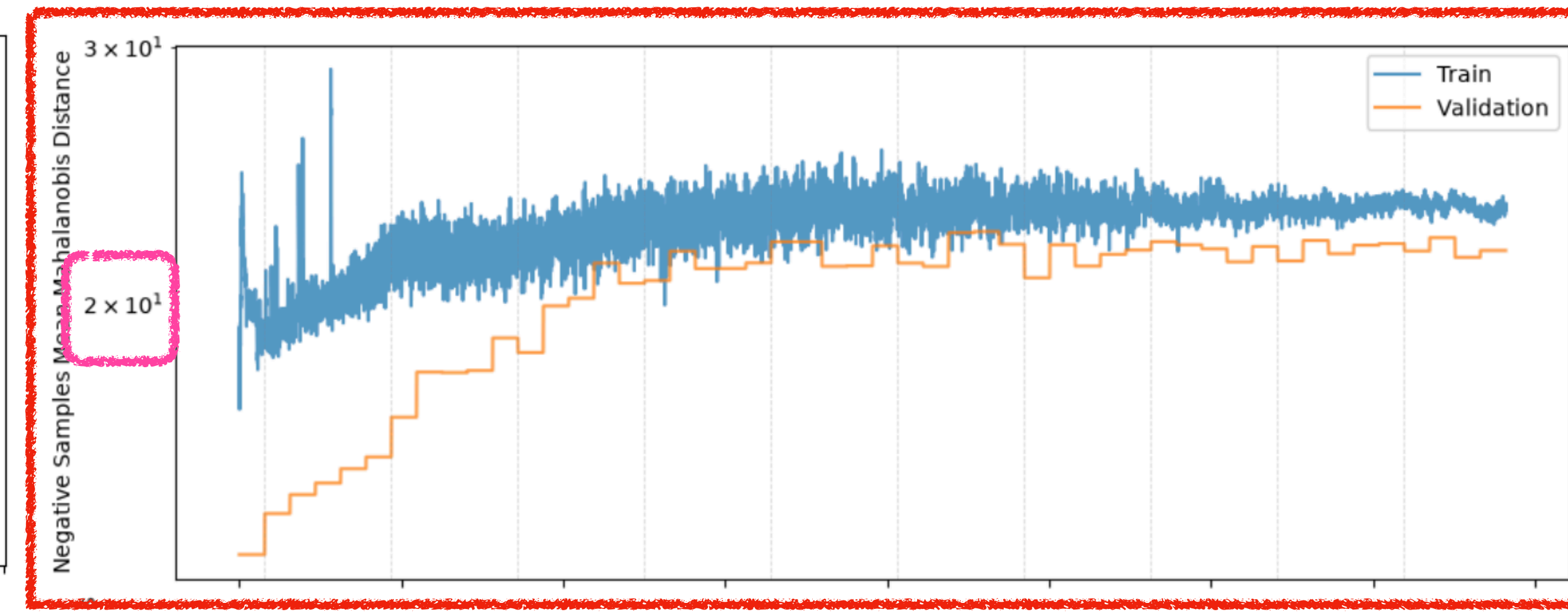
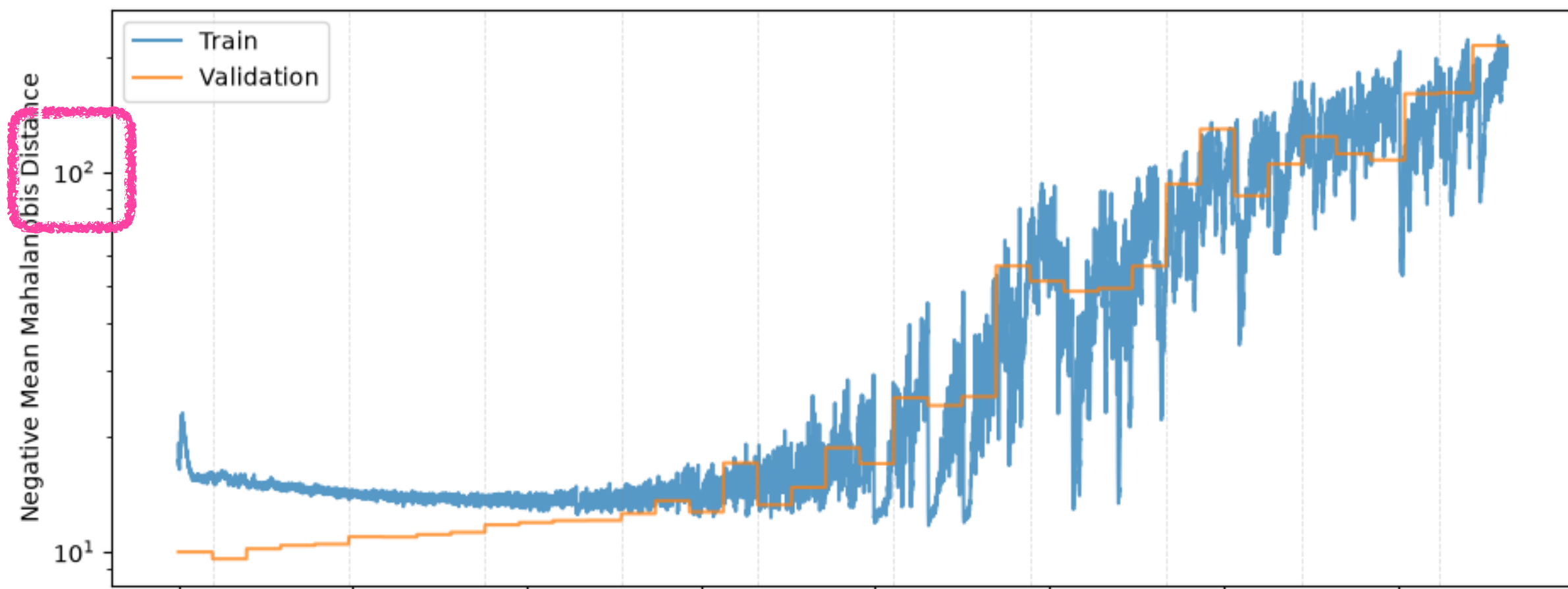
A negative result that clarified the role of triplet loss

- Without radius clipping, AUC slowly reached only **0.587**.
- With radius = 2, AUC converged around **0.583**, but Mahalanobis loss stayed almost zero.
- With radius scaled to 20, the model rapidly collapsed toward **0.5 AUC**, while Mahalanobis distance kept increasing.



No radius clipping. Model can reduce loss by infinitely pushing negative views away.

Radius = 2. Similar AUC dynamics, but all negative view latents are outside radius from the first step.



No radius clipping. Model can reduce loss by infinitely pushing negative views away.

Radius = 20. AUC degenerates. Negative views are pushed too far but not completely off the distribution, so that they distorted normal representation space perceivably.

Failed but useful experiment: Mahalanobis loss

A negative result that clarified the role of triplet loss

- Anchoring every negative views to train distribution center can make the model **overfit** to augmentation-specific artifacts or **mess up the causality** in representation space.
- Indeed, each corrupted view should be contrasted against the clean view of **the same event**.
- We returned to triplet loss, but more conservatively. It should act as a gentle regularizer, not the dominant training objective.

Transition to multi-background JetClass

From past CMS data to JetClass-scale

- Because the new CMS preprocessing took longer than expected, we used JetClass as a controlled large-scale testbed for multi-background anomaly detection.
- Jet types: **QCD, Hbb, Hcc, Hgg, Wqq**
- One class is held out as signal; others form the training background.
- Each jet type has **~10M** training events.
- With 40 epochs x 2000 steps x 128 batch size, each event is seen less than **0.25 times on average**.
- This is effectively stream-based learning rather than repeated finite-dataset training.

Transition to multi-background JetClass

Engineering contributions

- Implemented infinite-stream lazy dataset. ROOT shard requested on demand.
- Added shard shuffle and random event sampling; each data loader worker always holds all jet types.
- Implemented DistributedDataParallel (**DDP**) for multi-GPU to support larger effective batch size.
- Standardized read_file() API so JetClass and future CMS datasets can be swapped with ease.

Dataset Raw Features

- Available particle features include:
- Four-momentum: `part_px, part_py, part_pz, part_energy, part_pt`
- Relative geometry: `part_deta, part_dphi`
- Track impact parameters: `part_d0val, part_d0err, part_dzval, part_dzerr`
- Particle identity: `charge, isChargedHadron, isNeutralHadron, isPhoton, isElectron, isMuon`

Preprocessing

- Basic event cuts:
- Keep jets with `jet_pt > 0 and jet_energy > 0. and |jet_eta| < 2.5`
- Keep particles with `part_pt > 0 and part_energy > 0 and  $\Delta R < 0.8$ .`
- **Not all features should enter the backbone**
- The feature list is rich, but we do not necessarily want every available variable to enter hidden states.
- Too much low-level information can make the pretext task too easy for learning abstractions.

Preprocessing

- Four-momentum is used for only attention-bias construction, not passed into hidden states at all.
 - Absolute particle `part_eta` and `part_phi` are removed.
 - Only relative geometry preserved by passing `part_deta` and `part_dphi`.
 - Replaced raw `part_pt` with event-normalized `part_pt_fraction`.
 - Added batch-normalized `log(part_pt_fraction)`.
 - Converted raw `d0`, `dz`, `d0err`, `dzerr` into significance values, followed by batch normalization.
-
- **The CLS representation task should not be solvable by reconstructing the global jet picture from a few over-informative particle variables. This pipeline forces the model to rely more on event-level structure and less on direct kinematic shortcuts.**

Training Objective

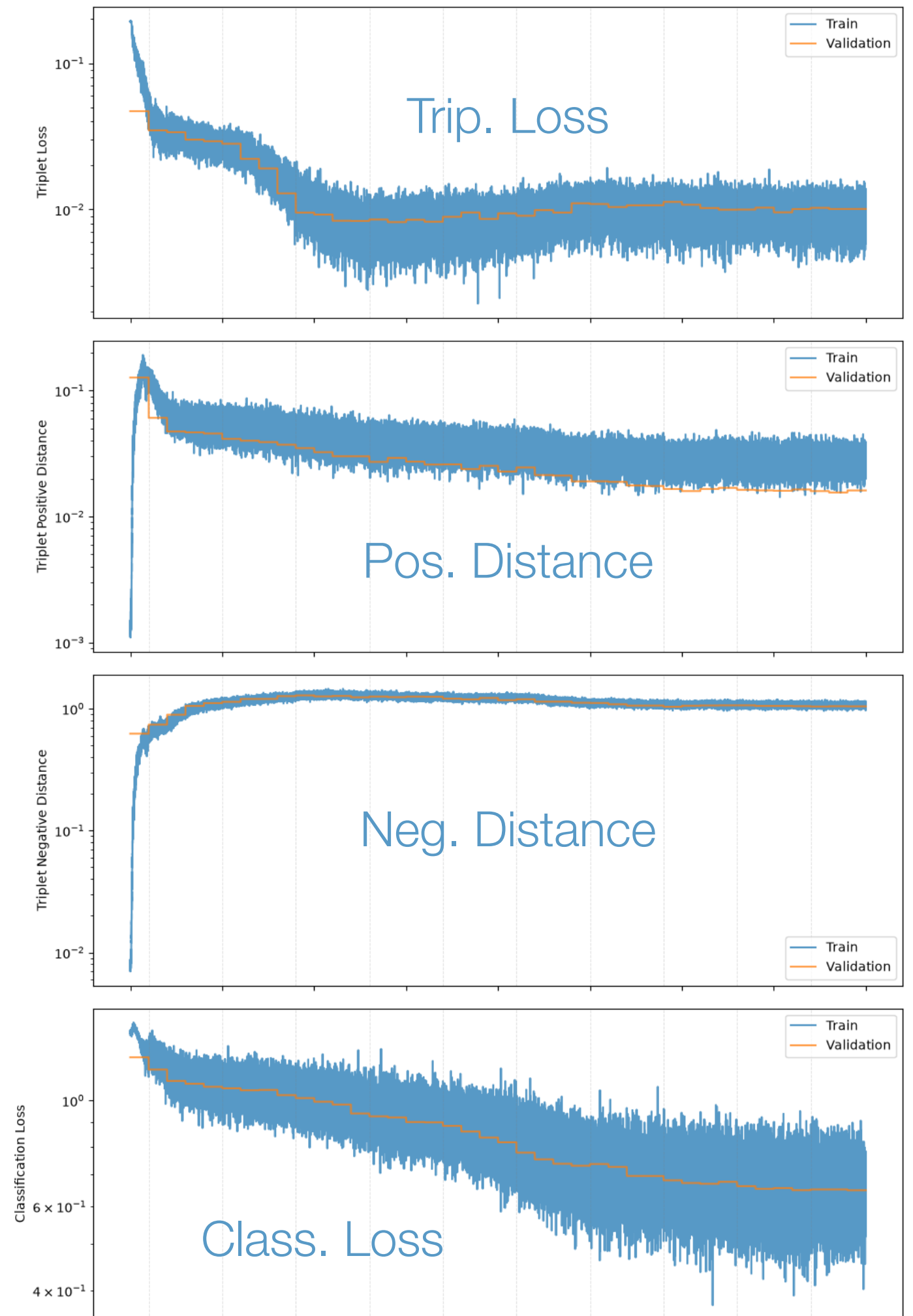
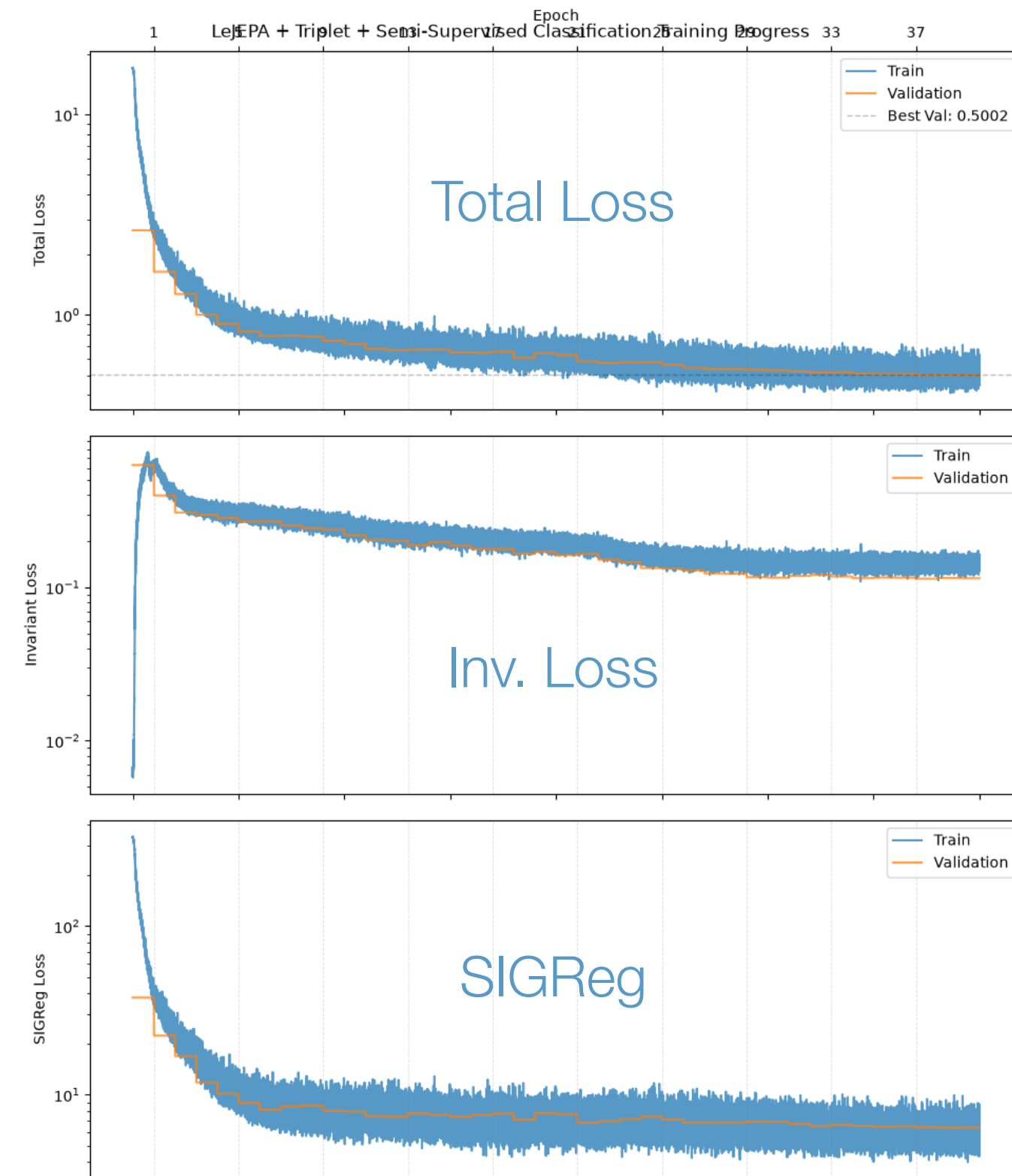
- Weakly-supervised multi-view representation learning with ParT backbone.
- Loss components:
 - **Invariant loss** aligns positive multi-view representations.
 - **SIGReg** prevents collapse and encourages isotropic gaussian distribution.
 - **Triplet loss** separates corrupted negative views from the corresponding clean event anchor
 - **Classification loss** predicts the background jet type during training
- Why classification:
 - The goal is not to train a supervised signal classifier. The classification head only uses training background labels, and it is used to encourage a more organized multi-background representation space.

Model Size

- Initial JetClass-scale runs became expansive. **100 epochs x 1000 steps** took a full day.
- Reduced model size:
- Hidden dimension: **128 -> 32**
- Representation dimension **128 -> 32**
- Transformer layers: **8 -> 4**
- Training: **40 epochs x 2000 steps x 128 batch size x (2 global + 4 local + 4 negative views)**
- Typical single L40S GPU runtime: **~5 hours**
- *Despite the much smaller model, the Wqq-as-signal model recovered QCD vs Wqq performance to the corrected single-background baseline.*

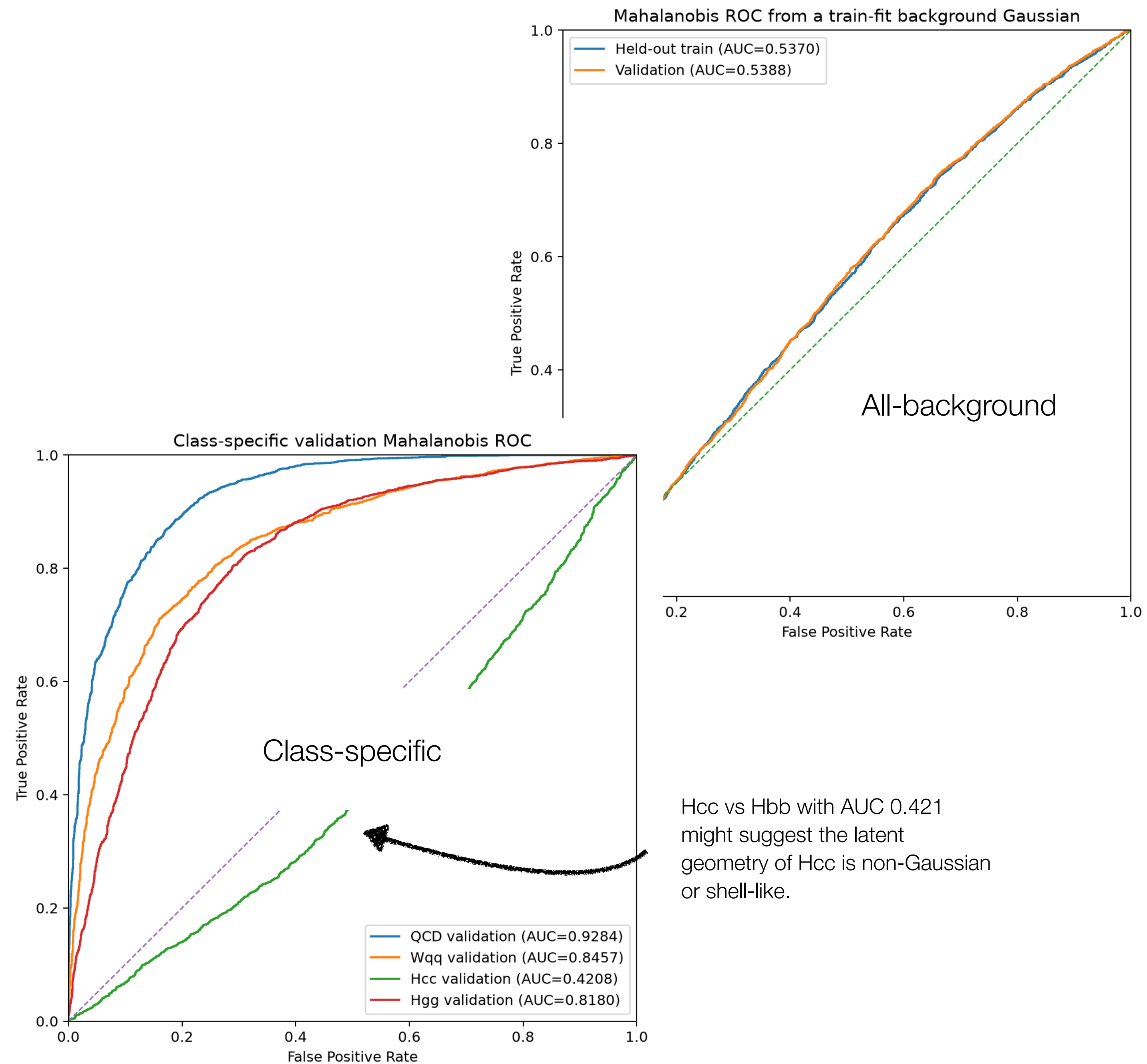
Hbb-as-signal

- Background: QCD + Wqq + Hcc + Hgg
 - Signal: Hbb
 - Triplet margin: 0.2
 - Epochs: 40
 - Min val loss: 0.5002
-
- Invariant loss steadily decreased.
 - Triplet loss did not collapse indefinitely.
 - This suggests the model learned view-invariant structure without simply overfitting to artificial augmentations



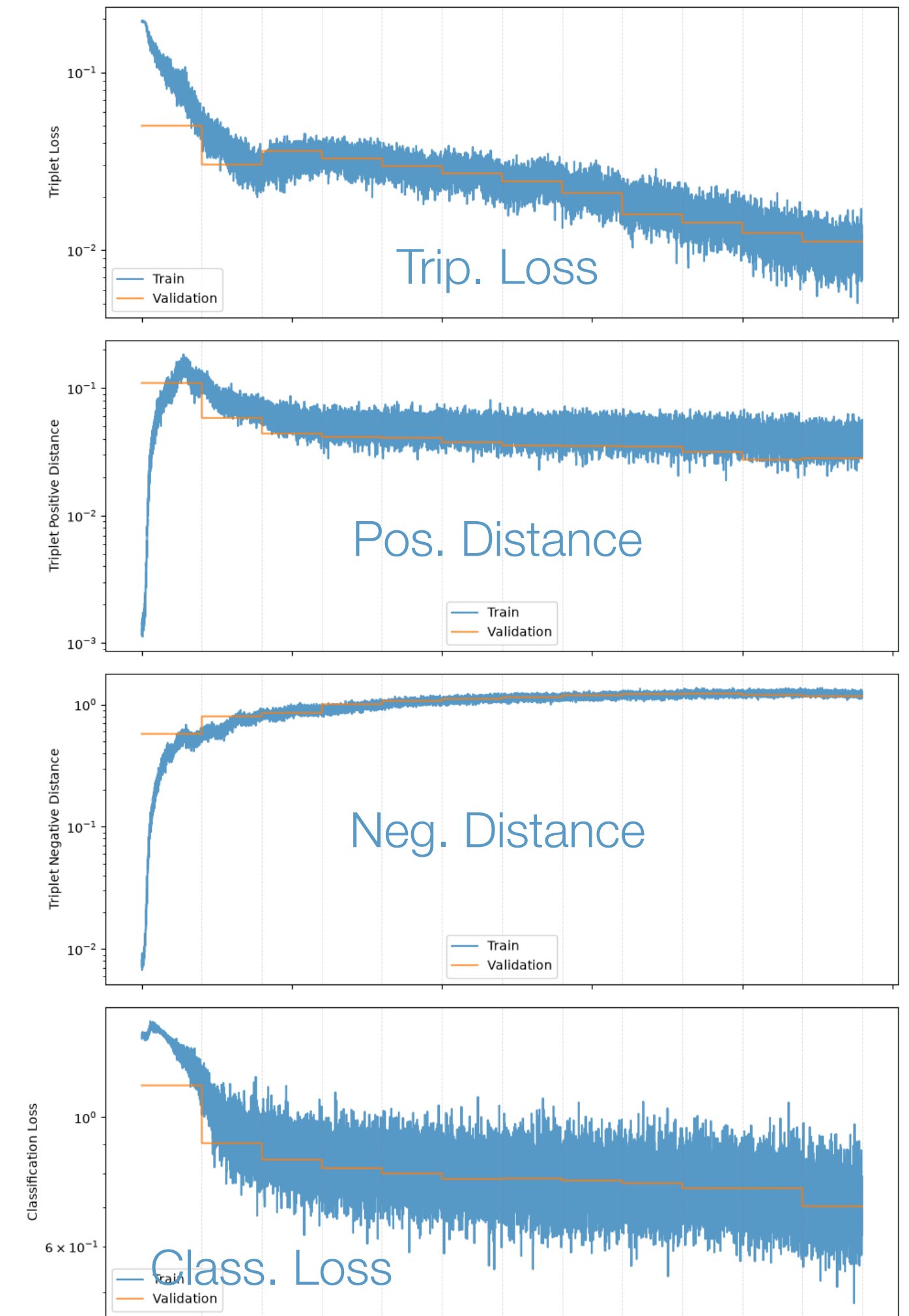
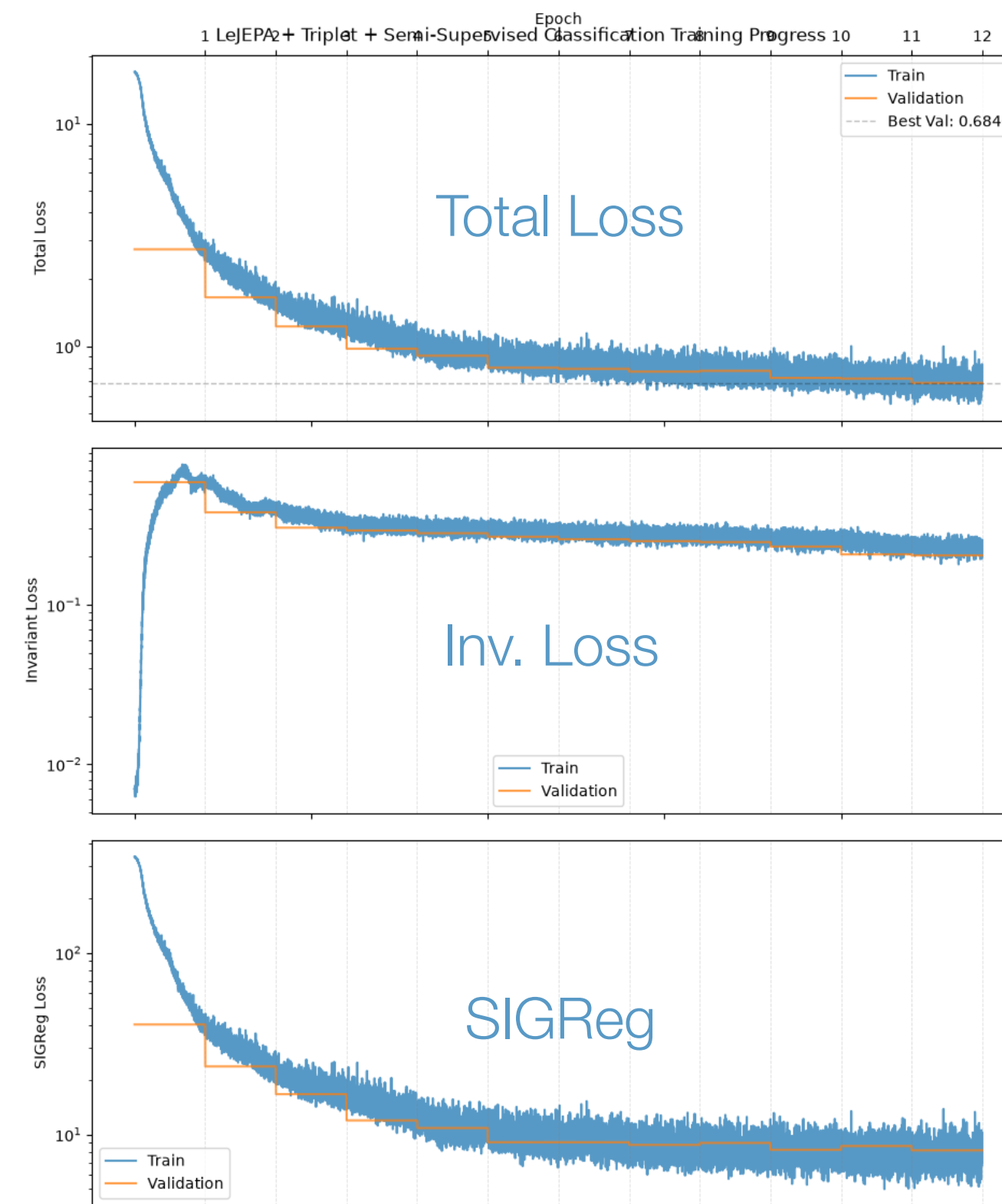
Hbb-as-signal

- **Per-class Mahalanobis ROC:**
- QCD vs Hbb: **0.928**
- Wqq vs Hbb: **0.846**
- Hgg vs Hbb: **0.818**
- Hcc vs Hbb: **0.421**
- All-background Mahalanobis: **0.539**
- The representation contains strong pairwise separation, but the merged background distribution is not well described by one Gaussian.



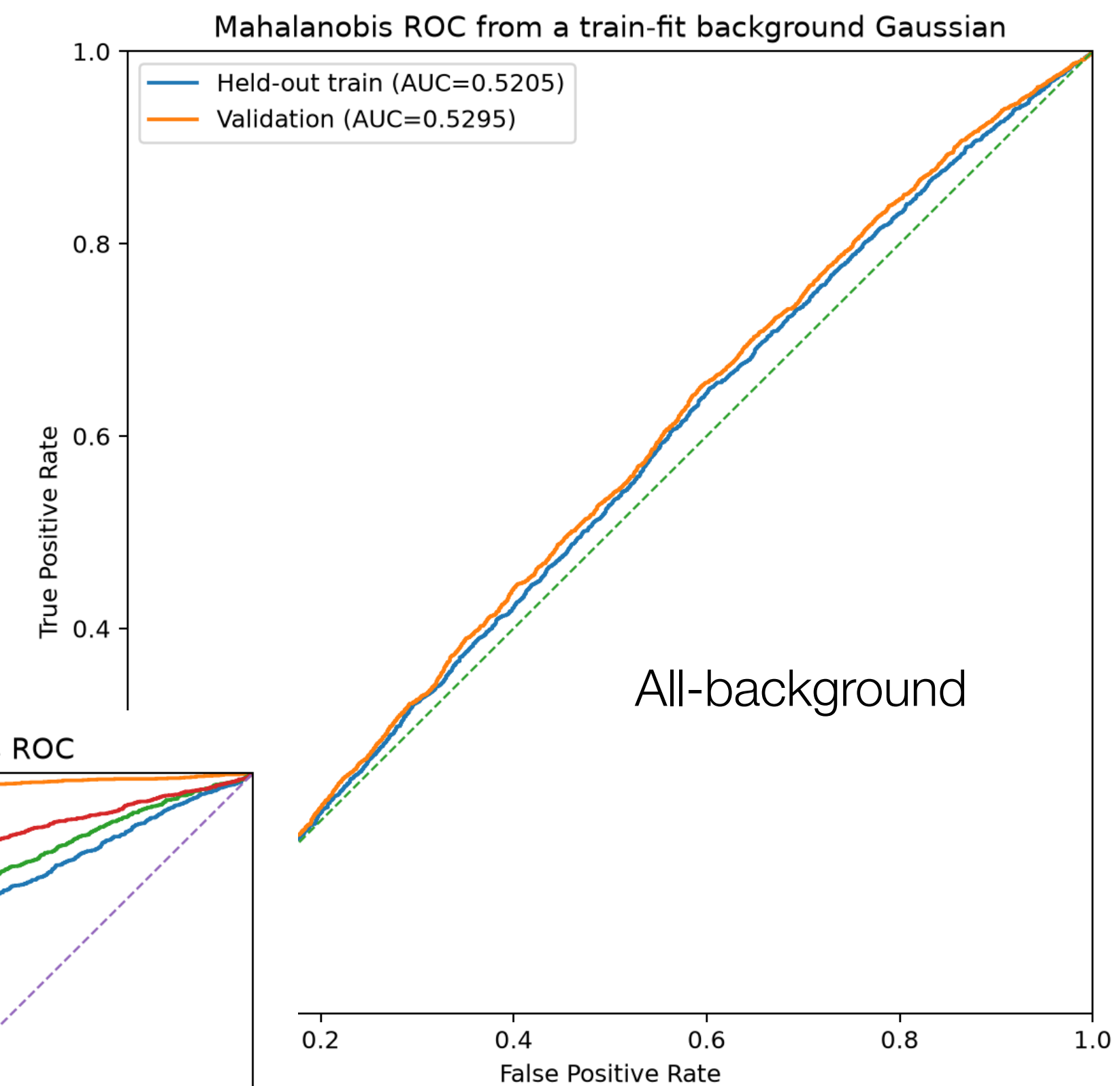
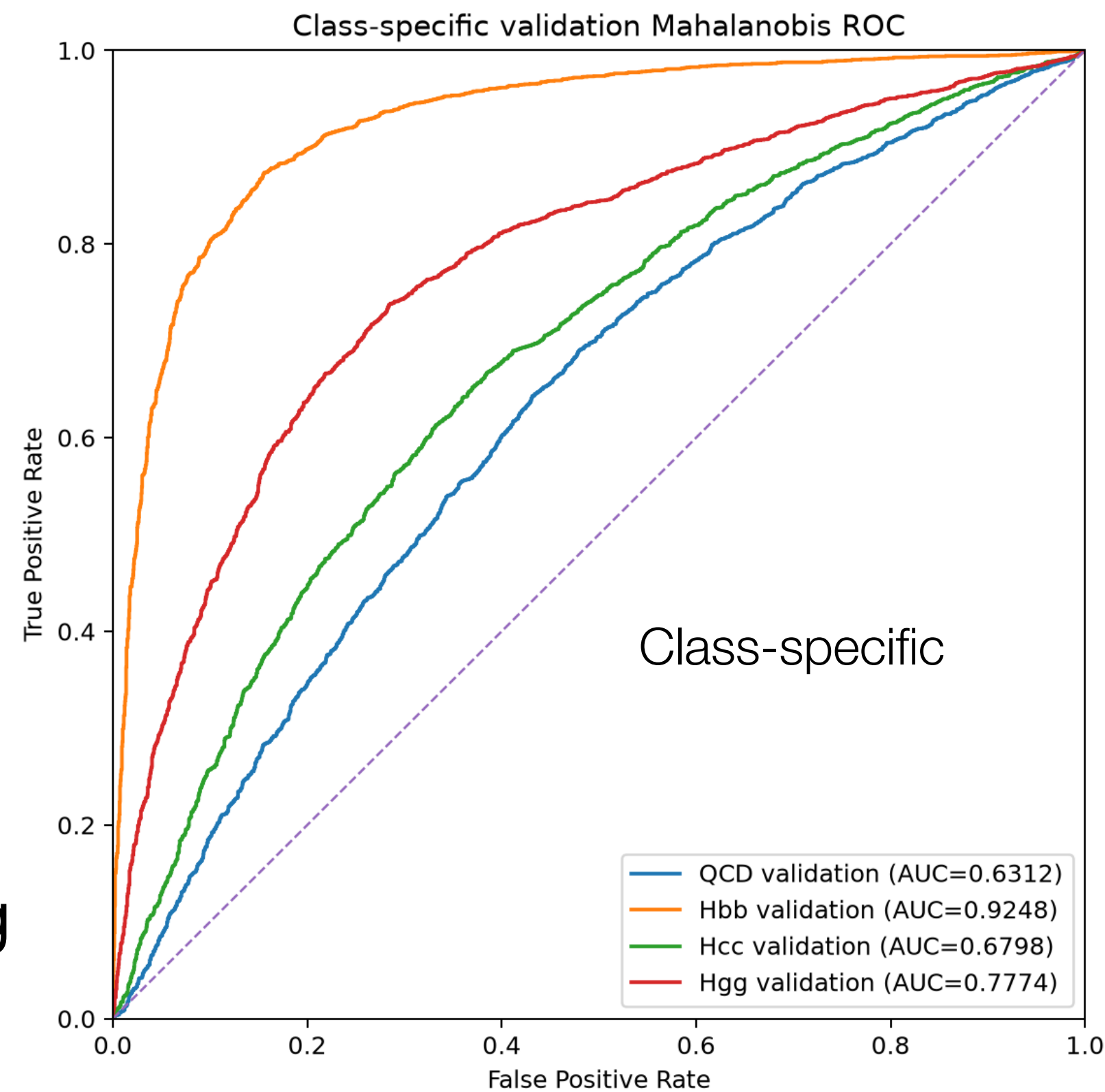
Wqq-as-signal

- Background: QCD + Hbb + Hcc + Hgg
- Signal: Wqq
- Triplet margin: 0.2
- Epochs: 12
- Min val loss: 0.6846
- Similar training dynamics.
- The script accidentally exited at 12 epochs



Wqq-as-signal

- **Per-class Mahalanobis ROC:**
- QCD vs Wqq: **0.631**
- Hbb vs Wqq: **0.925**
- Hgg vs Wqq: **0.680**
- Hcc vs Wqq: **0.777**
- All-background Mahalanobis: **0.530**
- The **QCD vs Wqq AUC of 0.631** is close to the corrected single-background baseline of **0.632**. Even with much smaller model, multi-background training did not erase the QCD–Wqq separation



Score Comparison

Anomaly Score	How it is computed	Observation
Global Mahalanobis	Fit one Gaussian to all validation background latents, then score by full-covariance Mahalanobis distance.	Fails in multi-background setting; Hbb AUC drops to 0.539 , Wqq AUC to 0.530
Per-class Mahalanobis	Fit a separate Gaussian for each background class, then compare signal against each class-specific normal region.	Strong pairwise separation, e.g. QCD vs Hbb 0.928 and Hbb vs Wqq 0.925 .
Triplet hinge violation	Recompute whether negative views remain closer to the anchor than allowed by the triplet margin: large violation means the event does not satisfy the learned clean-vs-corrupt separation.	Little discrimination.
Triplet margin score	Use the positive-negative distance gap induced by the triplet objective as an event-level score; larger score means the corrupted views are less cleanly separated from the positive anchor.	Useful for Hbb, around 0.661 AUC , but not robust for Wqq.
Local-global MSE	Compare local-view representation against global/full-view representation using MSE.	Little discrimination.
Local-global cosine	Compare local and global representations using cosine distance.	Little discrimination.
kNN density	Score events by distance to nearest validation-background neighbors in latent space.	Encodes useful local density changes; Hbb around 0.615 , Wqq around 0.595 .
Latent norm	Use the full-jet latent vector norm as a simple geometry diagnostic.	Useful for Wqq, little discrimination for Hbb

Node Features and Latent Quality

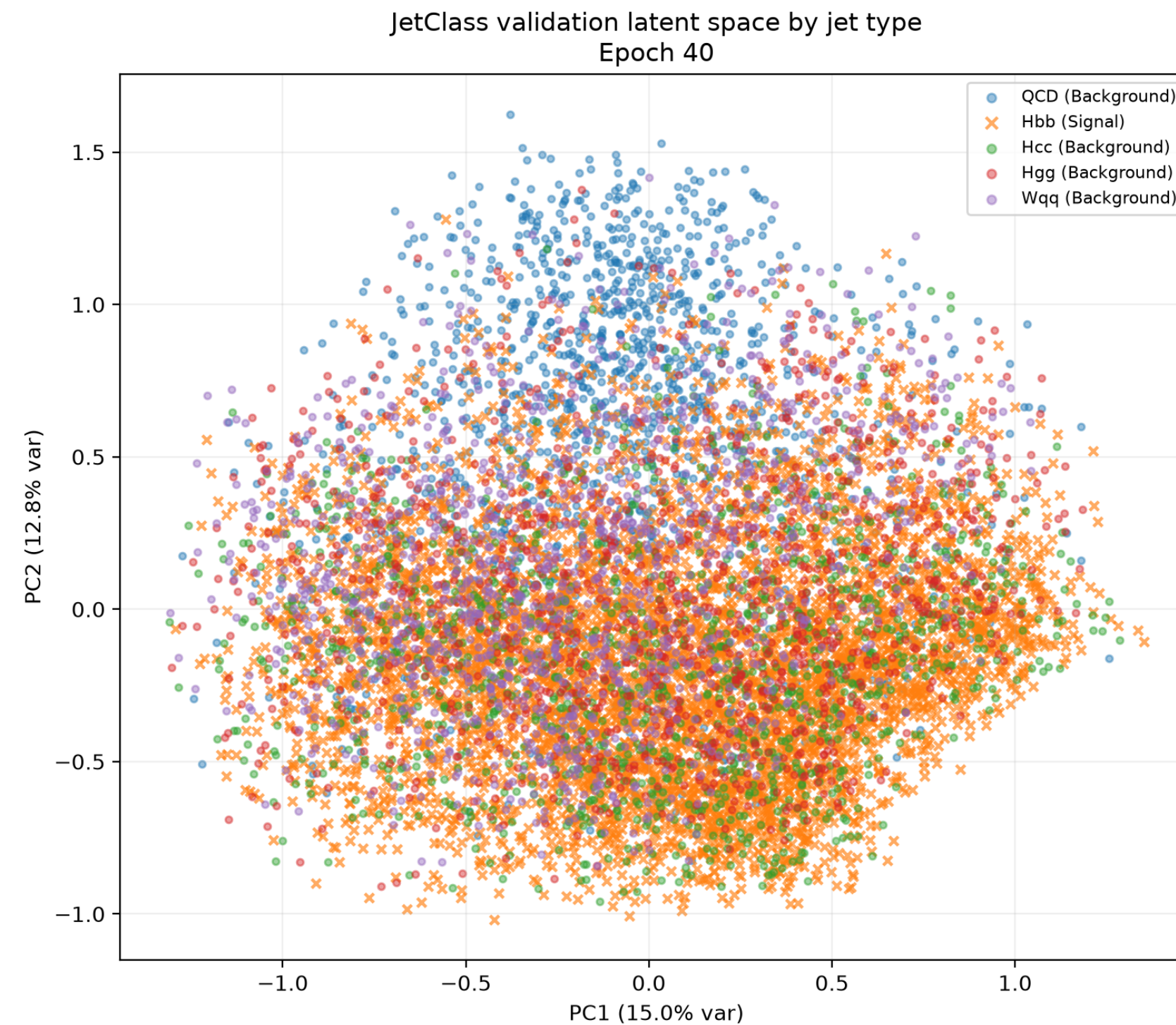
Not all features should enter the backbone

- The initial preprocessing pipeline fed extra information to backbone hidden states such as:
- Absolute particle `part_eta \ part_phi` coordinates
- Four-momentum
- Raw `d0, dz, d0err, dzerr` values
- No `log(part_pt_fraction)`

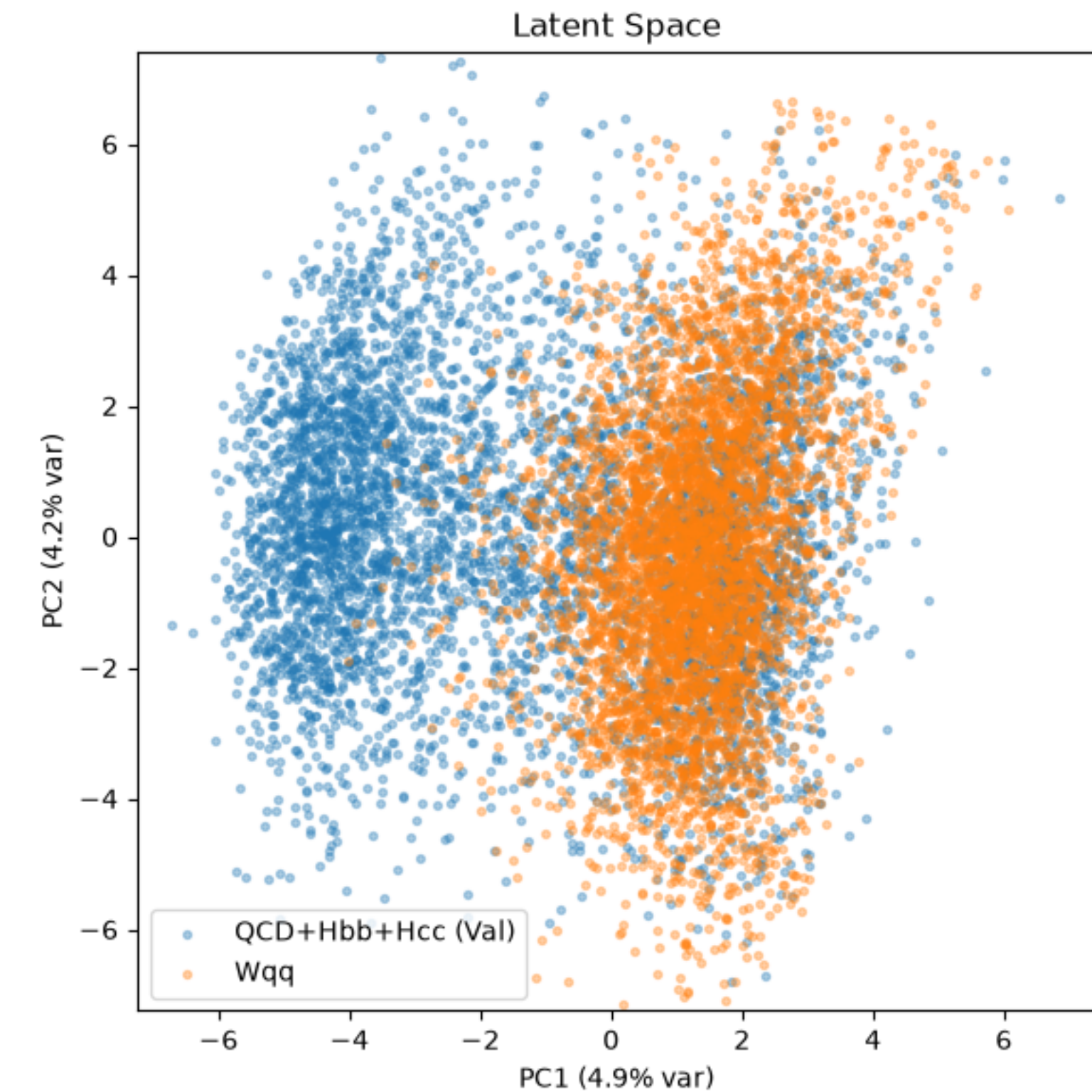
Node Features and Latent Quality

Not all features should enter the backbone

- Before fixing the pipeline, the latent space showed apparent clustering, but not a good LeJEPA abstraction.
 - PC1 and 2 explained only 4.9% and 4.2% variance
 - The model likely relied on shortcuts from over-informative variables
- With the improved pipeline, the LeJEPA latent space becomes much closer to a **unified isotropic Gaussian-like cloud**.
 - Jet types remain entangled in all-PCA (see pairwise PCA next)
 - Global structure smooth and compact
 - PC1 and 2 explain **15.0%** and **12.8%** variance.
- This suggests the representation has more organized dominant axes



Fixed preprocessing

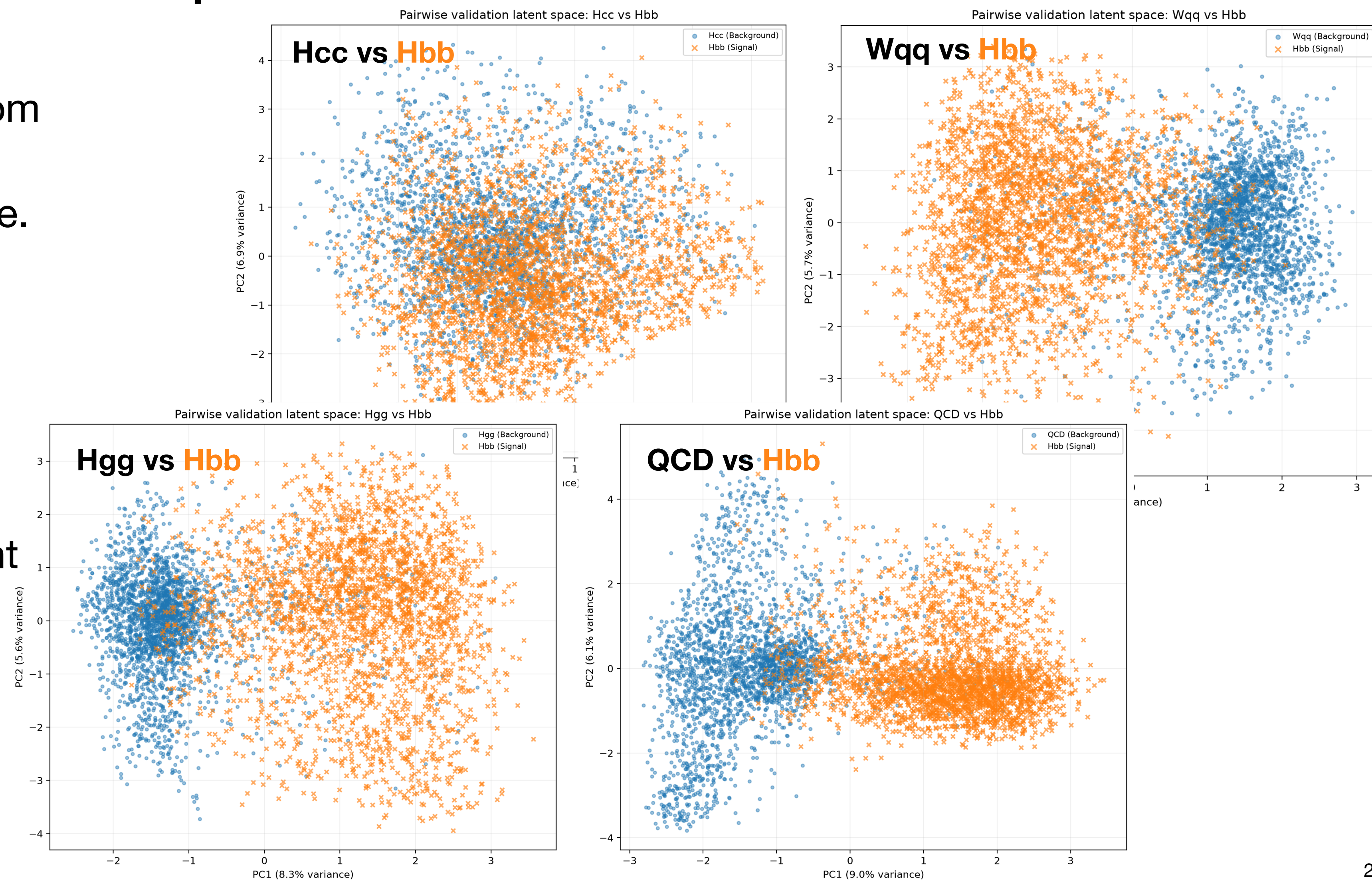


Initial preprocessing

Node Features and Latent Quality

Pairwise Jet Latent PCA for Fixed Pipeline

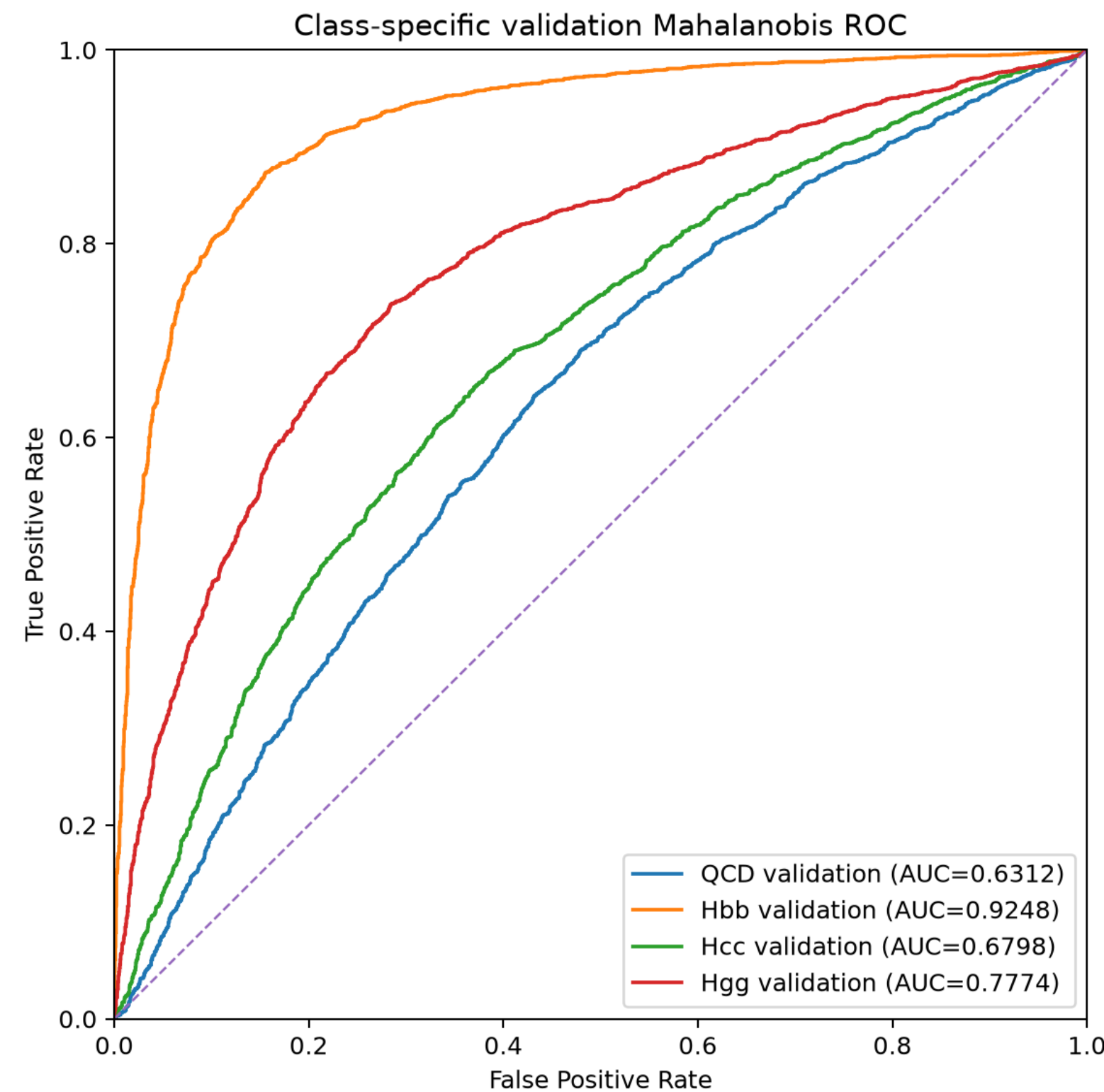
- We cannot separate signal from backgrounds if we view backgrounds at the same time.
- Pairwise latent PCA showed strong separation between backgrounds and the signal.
- Note the PCA axes are all different. Because different pairs are separated in different directions, their differences cancel each other out in all-PCA.



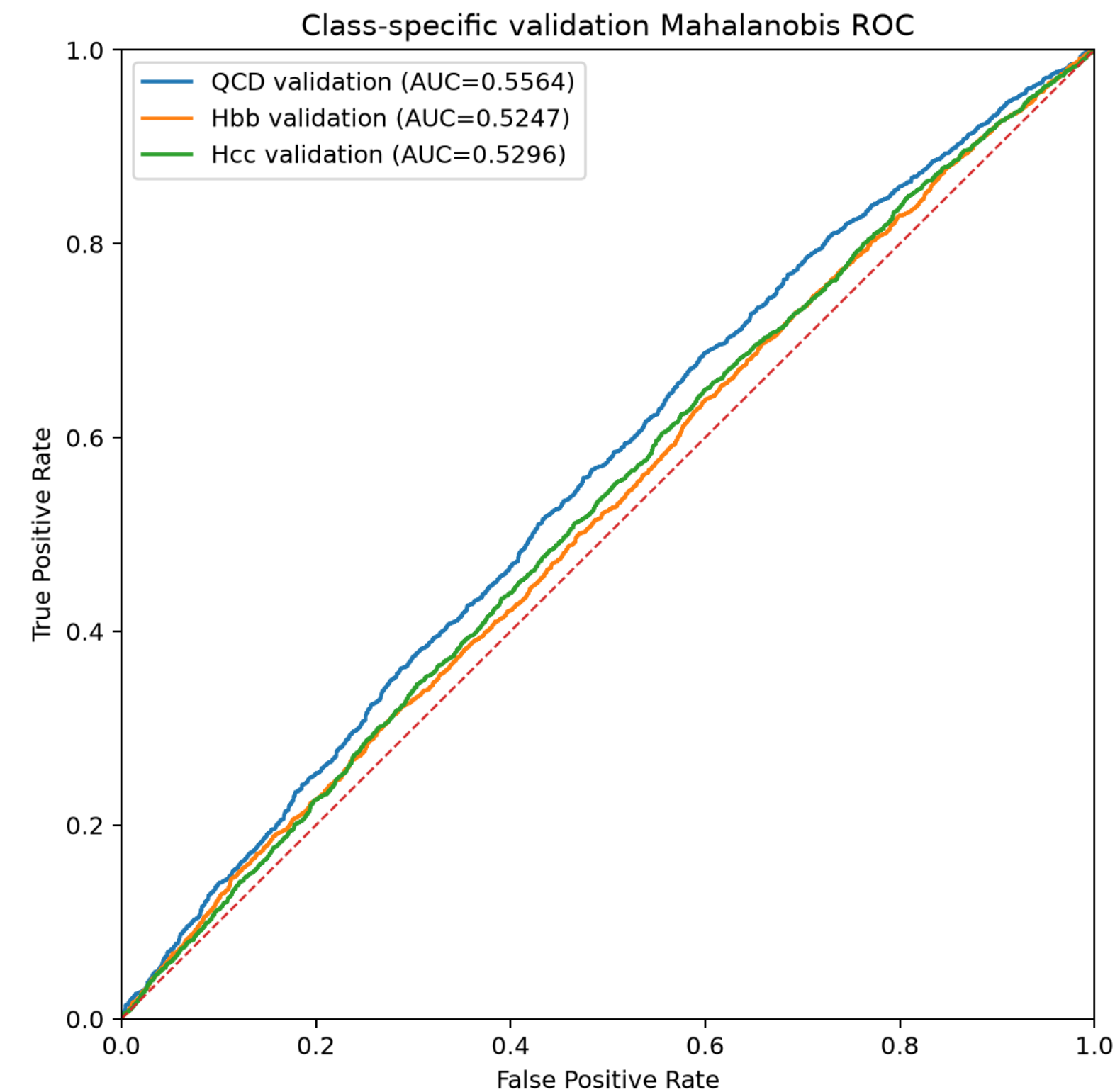
Node Features and Latent Quality

Not all features should enter the backbone

- The cleaner Gaussian-like structure for fixed pipeline yields significantly **more meaningful representation space**.



Fixed preprocessing



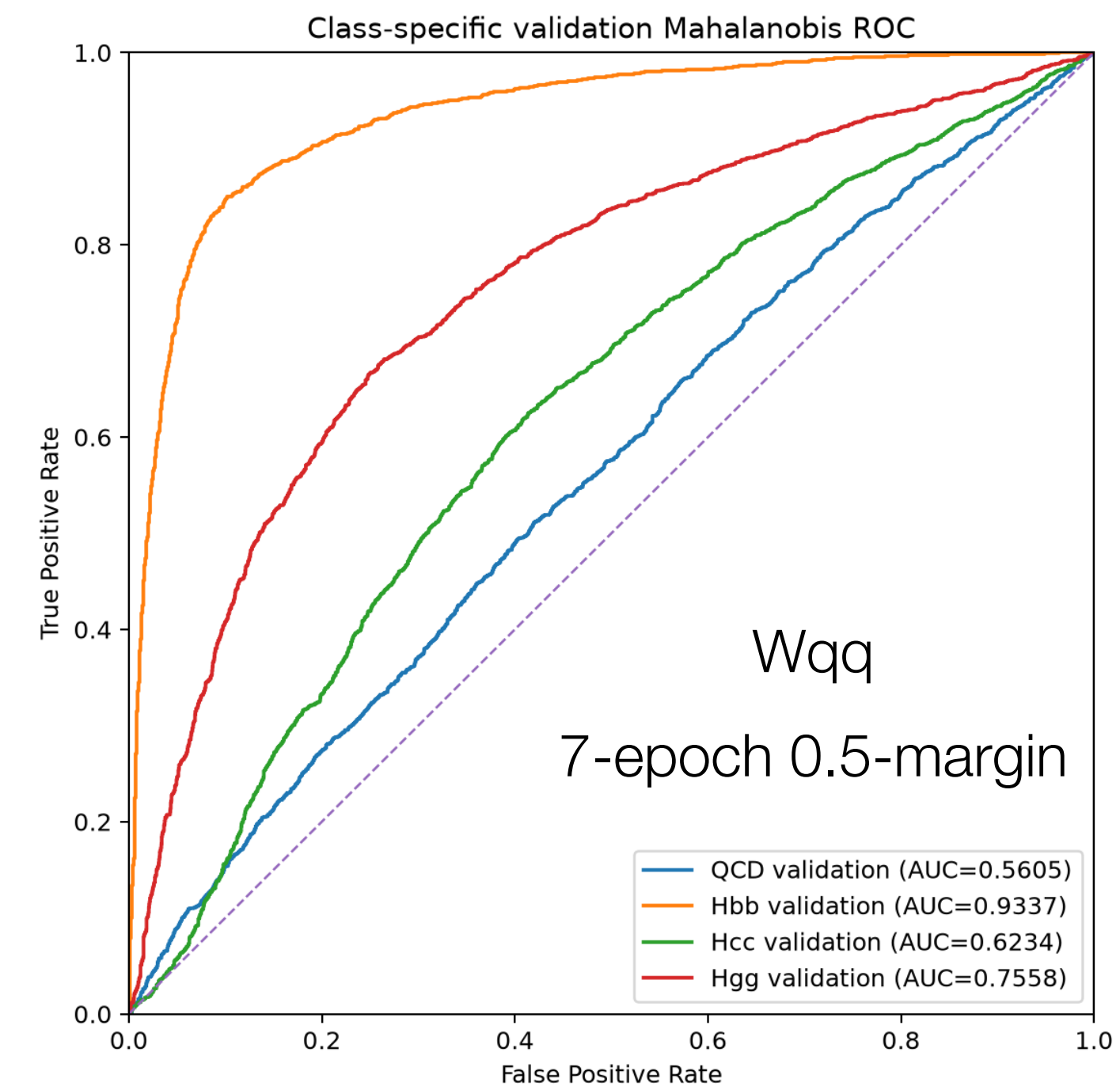
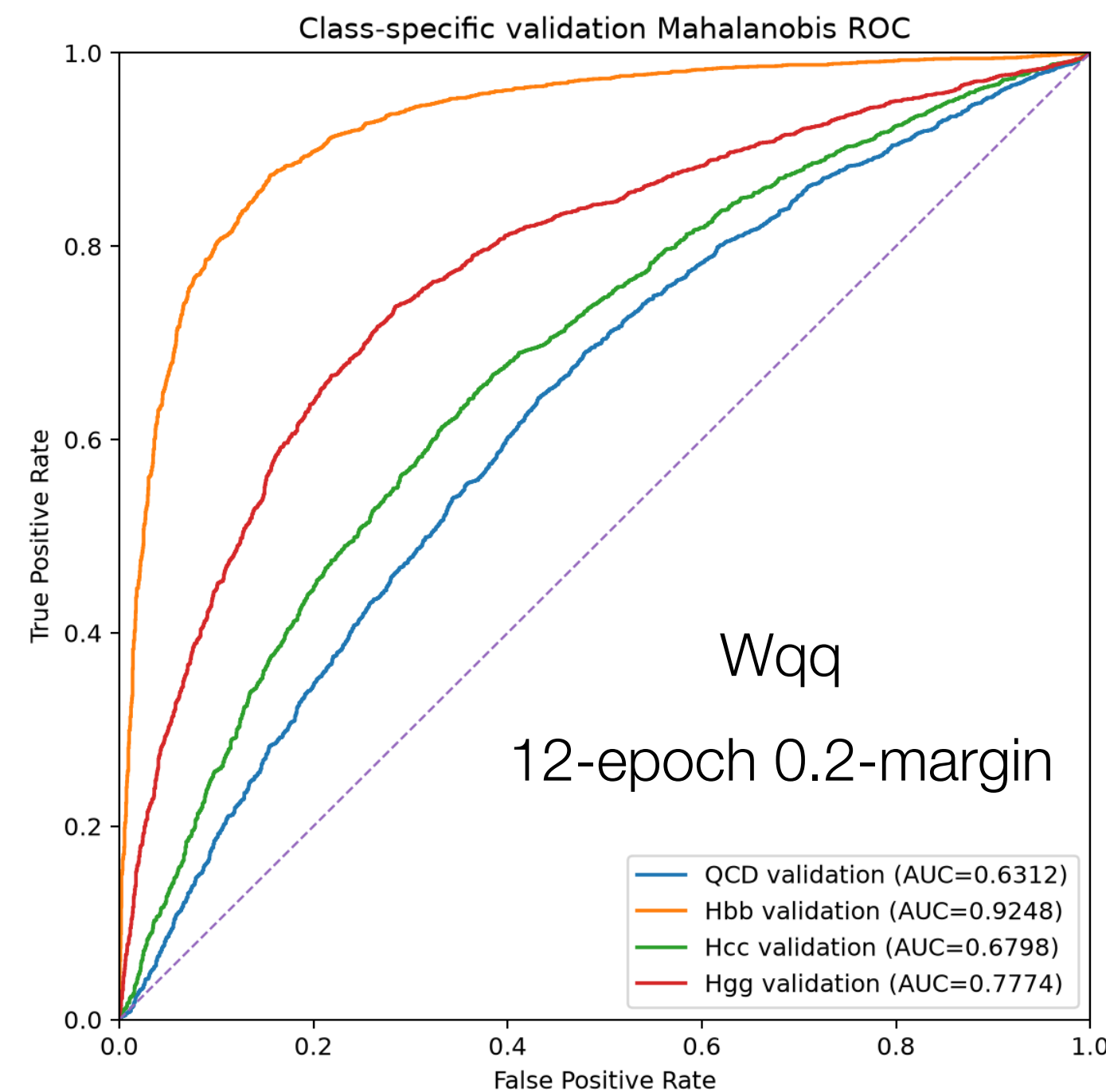
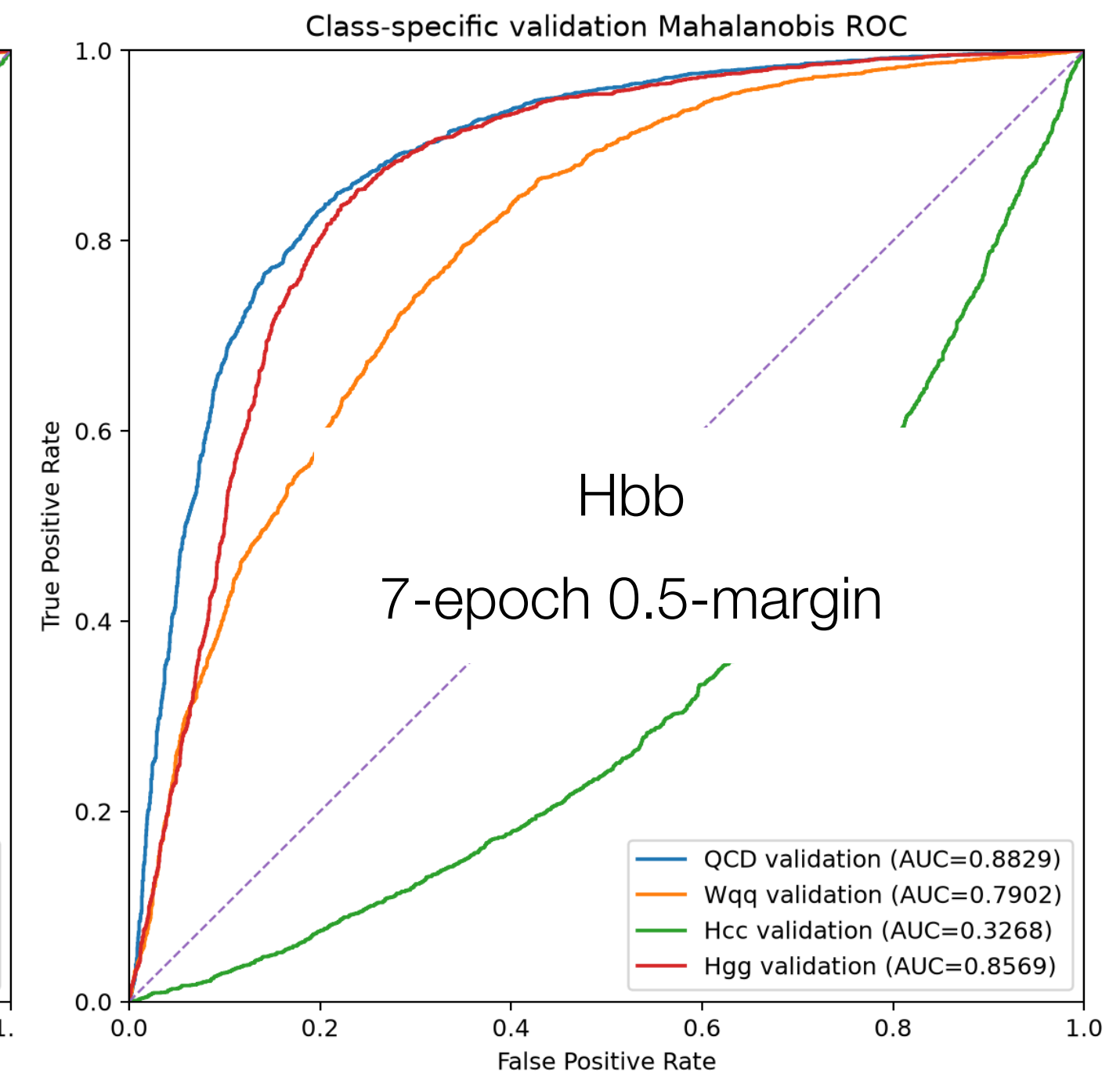
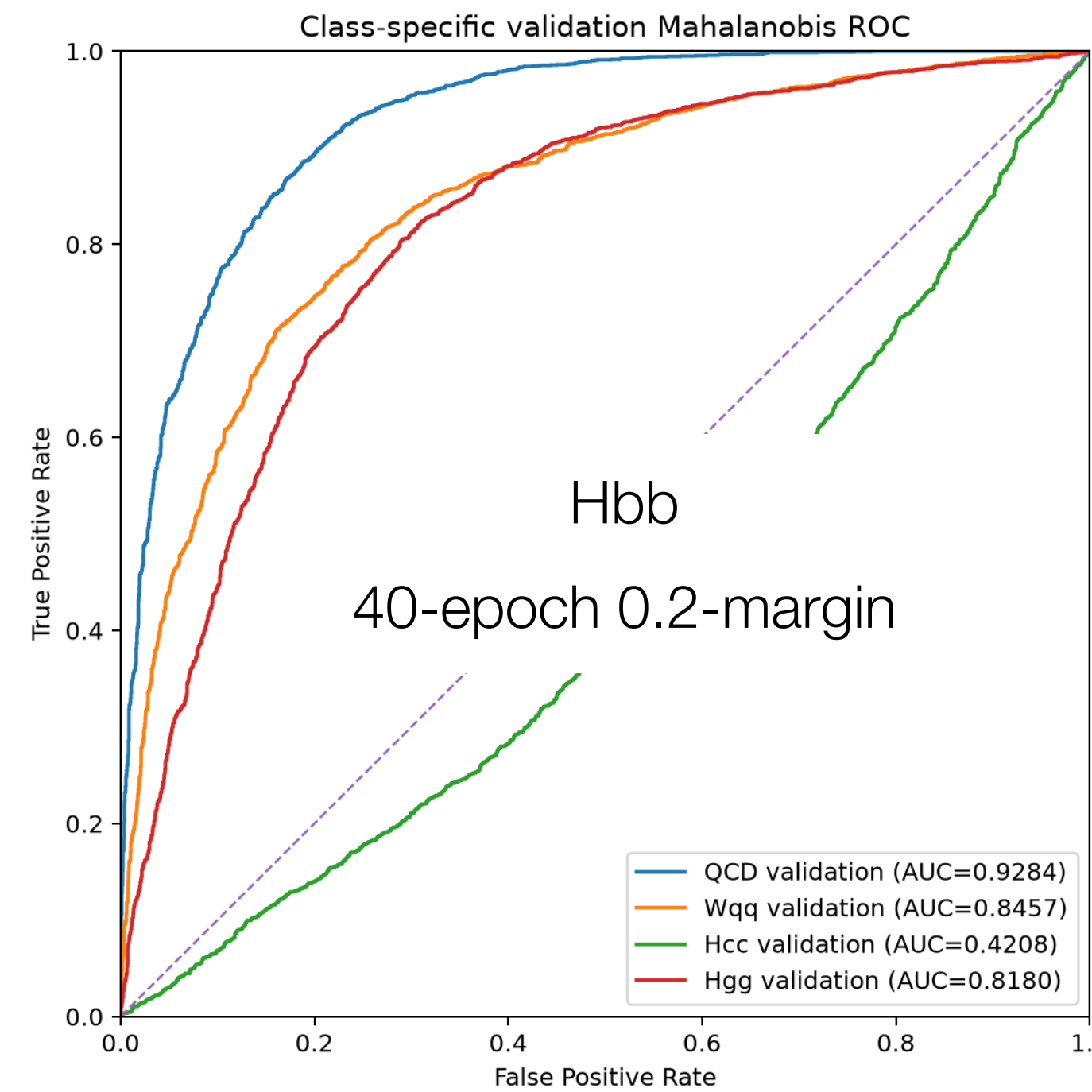
Initial preprocessing

Margin Study

- We also ran 7-epoch short experiments with triplet margin **0.5**, compared against the main **0.2-margin** runs.
- **Wqq model**
- The 12-epoch 0.2-margin model was better for **QCD, Hcc, and Hgg**. While Hbb vs Wqq was slightly better in the 7-epoch 0.5-margin model.
- **Hbb model**
- The 40-epoch 0.2-margin model was better for **QCD and Wqq**, while Hcc and Hgg favored the shorter 0.5-margin run.

Margin Study

- We cannot yet fully separate the effect of training length from triplet margin.
- The single-background studies showed that the margin should not be too large, because large margins can make the model overfit to artificial augmentation patterns.
- However, the multi-background results could also suggest margin **0.2** may be too weak to provide enough separation pressure compared with **0.5**.
- We need a controlled margin sweep under fixed training length in the future.



Scaling Up

Moving on from PFNano and JetClass

- Ntuplizing our own ROOT files from MiniAOD's.
- Filtering explicitly for AK8 jets and acquiring more known anomalies for training.

Samples & scale			
Class	Datasets	MiniAOD files	Selected jets
QCD	7 pT bins (170→2000 GeV)	1400	(bulk of ~143M)
W	Wto2Q PTQQ 200/400/600	600	~12M / bin-scale
Z	Zto2Q PTQQ 200/400/600	600	~10M / bin-scale
ttbar	TTto4Q	200	~4M
Hbb	leptonic WH/ZH only	189	~1.1M total (~75k fj_isH)
Total		~2990	~144M

Selection (identical for all samples)	
Cut	Value
Jet pT	150–2000 GeV
η	≤ 2.5
Jet radius	AK8 (R=0.8)
FatJet info	ON (runFatJet=True)
gluonReduction	0 (no thinning)
PF cand pT	≥ 0.1 GeV

Scaling Up

Details about branches/columns for each jet:

2. Charged PF (Cpfcan_*) — particle cloud

Kinematics: pt, eta, phi, px/py/pz, e, p_{rel}, e_{rel},
etarel, phirel, deltaR

ID / weight: charge, pdg / pdgID, isMu, isEl, puppiw,
HadFrac, CaloFrac

Track / IP: dxy, dxyerrinv, dxysig, dz, SIP2D/3D
(BtagPf_trackSip*), chi2, quality, fromPV,
lostInnerHits, numberOfPixelHits, numberOfStripHits
(+ more b-tag track vars: decay length, PPar, PtRel, jet
distance, ...)

3. Neutral PF (Npfcan_*)

Kinematics: pt, eta, phi, px/py/pz, e, p_{rel}, e_{rel},
etarel, phirel, deltaR

ID: puppiw, pdgID, isGamma, HadFrac, CaloFrac
(+ drminsv, tau_signal)

1. Jet-level kinematics / IDs

- jet_pt, jet_eta, jet_phi, jet_mass, jet_energy
- jet_corr_pt, jet_qgl, jet_jetId / jet_looseId, jet_puId
- Flavour: jet_hflav, jet_pflav, jet_phflav, jet_pflavCharge
- Matches: jet_genmatch_pt, jet_genmatch_wnu_pt, jet_mumatch_pt, jet_elematch_pt, jet_taumatch_pt
- **Jet charge:** jet_qk_charge_05, jet_qk_charge_10
- Counts: n_Cpfcand, n_Npfcand (+ n_LTcand)

4. FatJet / substructure (fj_*) — key for anomaly / Hbb

Kinematics: fj_pt, fj_eta, fj_phi, fj_mass

Soft-drop / N-subjettiness: fj_sdmass, fj_tau1/2/3, fj_tau21, fj_tau32, fj_sdn2

Soft-drop subjects (sj1/sj2): pt/ η / ϕ /mass, CSV/JP, ptD, axes, multiplicity

Truth labels: fj_isH, fj_isW, fj_isTop, fj_isLight (+ fj_gen_pt/eta)

Double-b style inputs: fj_doubleb, fj_z_ratio, track SIP vars, tau-vertex vars, ...

(Hbb preprocess: keep fj_isH==1 $\rightarrow$ both truth b's inside $\Delta R < 0.8$)

- **Secondary vertices** sv_* (pt/mass/flight distance/sig, ...)
- **DeepFlavour scores** (probb, probbb, probc, probg, problepb, probuds)
- **CSV TagVars**, lost tracks (LT_*), V0/Ks, gen particles, flavour flags (isB/C/G/...), event meta (event_no, npv, rho, ...)

Takeaways - Main Results

- We scaled the framework from single-background QCD to JetClass-scale multi-background training
- With a much smaller model, we recovered QCD vs Wqq performance to the single-background baseline.
- Hbb and Wqq models both show strong pairwise separation under **per-class Mahalanobis scores**.
- The all-background Mahalanobis score can fail, as the background latent distribution is multi-modal.
- Multi-background anomaly detection should not be evaluated only with a single global normal distribution.
- Feature design matters: more variables can make the pretext task easier but the representation worse.
- Some failures, especially Hcc vs Hbb, suggest that certain background classes may have **non-Gaussian or shell-like** latent geometry.

Next Steps

- Continue training the current small model for longer.
- Run a controlled triplet-margin sweep: 0.1, 0.2, 0.5, 1.0
- Further reduce node features to make view matching harder
- Remove impact parameters / particle identity
- Investigate why Hcc vs Hbb behaves anomalously. Check Hcc latent geometry.
- Port the new CMS dataset with `read_file()` interface.
- Rerun the same training once preprocessing is ready.